MSc. in Mechanical Engineering

MECA0029-1| Theory of Vibration

Project Report
Analysis of the dynamic behavior of a steel canopy

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Context

This project focuses on analyzing the dynamic behavior of the steel canopy shown in Figure 1.
The first part of the analysis will focus on developing the finite element model on python. The accuracy of the model will then be assessed by performing a modal analysis, the results of which will be compared and validated against the results obtained from the same analysis conducted in the commercial software Siemens NX, taken as reference.
The stationary and transient responses will then be studied, as well as the reduction methods, applied to the python model.

Figure 1: Picture of the steel canopy studied

1 Modeling of the structure

Model

For the analysis, the steel canopy is modeled by "two frames, spanning 15 m in the X-direction, in red in Figure 2, with support beams forming a zig-zag pattern every 1.5 m in the X-direction. The two frames are 4 m apart in the Y -direction and are linked by transverse support beams. Support and transverse beams are depicted in blue in the Figure 2." The four ends located on the lines X = 0 and X = 15 m are clamped.
The beam properties used in the model are the following :

- Density: $\rho = 7800$ kg/m^3
- Poisson's ratio: $\nu = 0.3$
- Young Modulus: $E = 210$ GPa
- Circular hollow sections :
  - Frame (red) beams: outer diameter of 120 mm and wall thickness 5 mm
  - Support and transverse (blue) beams: outer diameter of 70 mm and wall thickness 3 mm
- A lumped mass: $M = 500$ kg distributed on the 8 pink nodes (see Figure 2), representing the mass of the roof.
- Rotational inertia of the roof is neglected

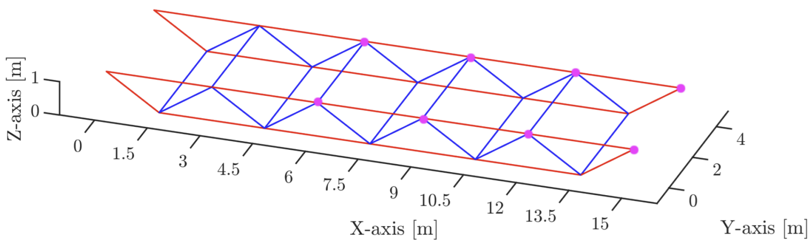


Figure 2: Model of the steel canopy used for the analysis of the dynamic behavior

Finite element modeling

A finite element approach is used for the further analyses. This approach consists in solving partial differential equations (PDE's) problems by discretizing the continuous system – and its equations– into smaller *elements* of given properties, in order to transform the PDE's into a linear system of equations, which can be solved using algebra.

For this structural dynamics problem consisting in a series of beams, the discretization is made by dividing the entire structure into smaller beam elements of given length l_e characterized, for their dynamic analysis, by *local* –identified by the bar on top of the variable– elementary stiffness and mass matrices $\overline{\mathbf{K}_e}$ and $\overline{\mathbf{M}_e}$ in element-local system of coordinates, allowing for a universal shape of these matrices.
These matrices are then globalized by multiplying them by a rotation matrix $\mathbf{R}_e$ to account for the orientation of each element with regards to the global system of axes, writing as

$$\begin{aligned} \mathbf{K}_e &= \mathbf{R}_e \cdot \overline{\mathbf{K}_e}; \\ \mathbf{M}_e &= \mathbf{R}_e \cdot \overline{\mathbf{M}_e}. \end{aligned} \tag{1.1}$$

From there, matrices are then assembled to form the global stiffness and mass matrices $\mathbf{K}_s$ and $\mathbf{M}_s$ by using each element's localization matrix $\mathbf{L}_e$ (it represents where each of the nodes of the element lies in the global structure) using

$$\begin{aligned} \mathbf{K}_s &= \sum_e \mathbf{L}_e^{\mathrm{T}} \mathbf{R}_e^{\mathrm{T}} \overline{\mathbf{K}_e} \mathbf{R}_e \mathbf{L}_e; \\ \mathbf{M}_s &= \sum_e \mathbf{L}_e^{\mathrm{T}} \mathbf{R}_e^{\mathrm{T}} \overline{\mathbf{M}_e} \mathbf{R}_e \mathbf{L}_e + \mathbf{M}_L \end{aligned} \tag{1.2}$$

where $\mathbf{M}_L$ is the matrix of lumped masses, a diagonal matrix with non-zero terms equal to m_L on all the translational DOFs of the nodes subject to a given lumped mass m_L.

The constrained degrees of freedom (DOFs) from the fixed nodes are dealt with by partitioning the displacement vector $\mathbf{q}$ –the solution– and the global matrices into

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_R \\ \mathbf{q}_G \end{bmatrix} = \begin{bmatrix} \mathbf{q}_R \\ \mathbf{0} \end{bmatrix} \tag{1.3}$$

$$\mathbf{K}_s = \begin{bmatrix} \mathbf{K}_{RR} & \mathbf{K}_{RG} \\ \mathbf{K}_{GR} & \mathbf{K}_{GG} \end{bmatrix}$$
$$\mathbf{M}_s = \begin{bmatrix} \mathbf{M}_{RR} & \mathbf{M}_{RG} \\ \mathbf{M}_{GR} & \mathbf{M}_{GG} \end{bmatrix} \tag{1.4}$$

where the indices R and G respectively stand for *reduced* and *given*.
Finally, the modal analysis is performed by solving the discretized eigenvalue problem (the free response of the system)

$$(\mathbf{K}_{RR} - \omega_{(i)}^2 \mathbf{M}_{RR})\mathbf{x}_{(i)} = 0 \tag{1.5}$$

on the reduced system, for the first six natural frequencies $\omega_{(i)}^2$ and their corresponding spatial displacement modes $\mathbf{x}_{(i)}$.

The discretization strategy used for the project is to divide all beams into an equal number of elements.
In the Python model, the elementary matrices are based on the Euler-Bernoulli beam theory whereas Siemens NX uses elementary matrices based on the Timoshenko beam theory[1] (see Appendix C for the four elementary matrices expressions). This is the only conceptual difference in between the python and Siemens NX implementations of the method.

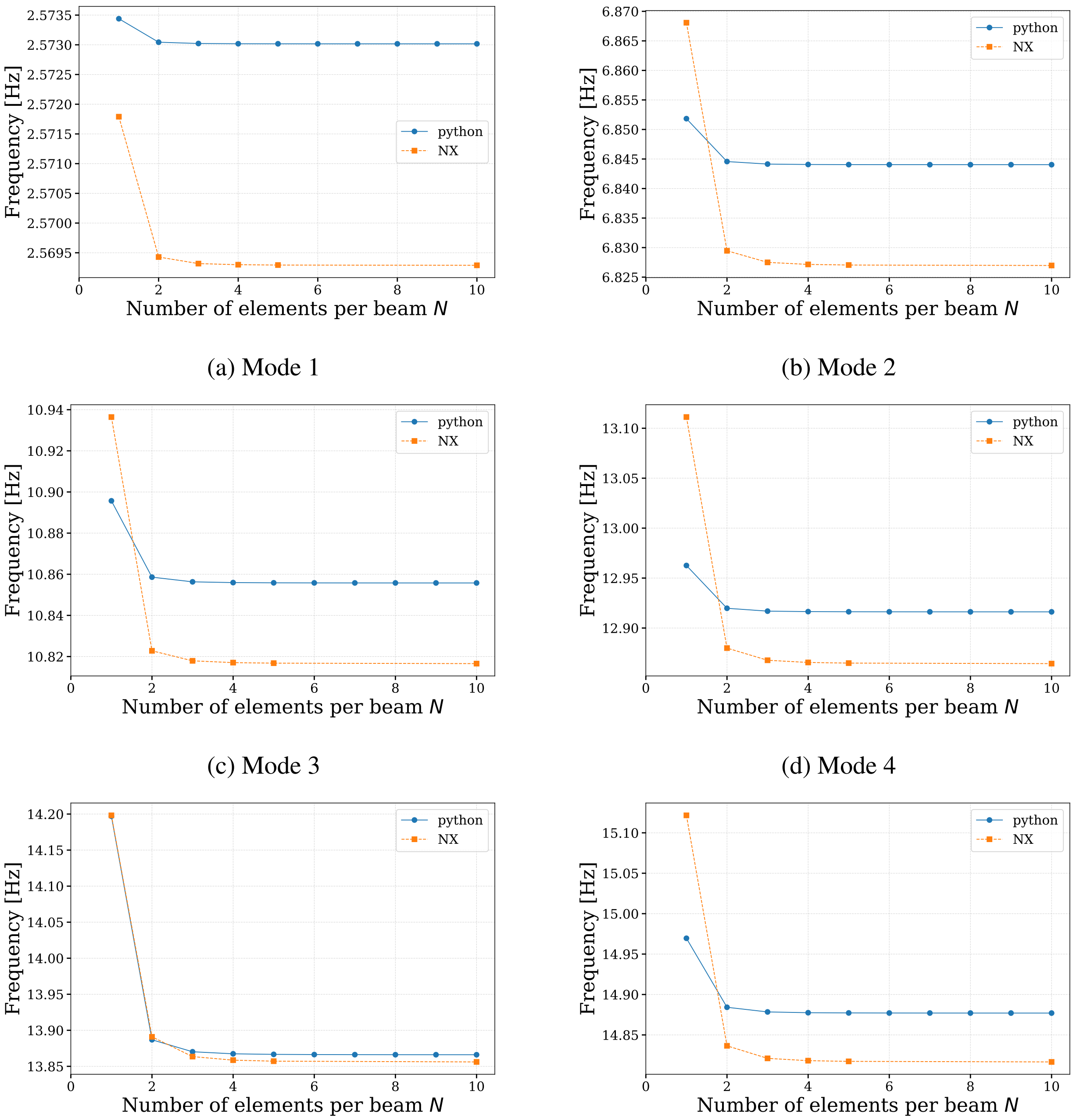
1.1 Convergence study

A mesh refinement study is first performed to assess the accuracy of the later results.
The refinement strategy consists in increasing, from one to ten, the number N of elements per beam in the model, this number being equal across all elements.
The convergence of the six first natural frequencies as a function of N, for both python and Siemens NX models, is presented in Figure 3. The relative change of the frequencies in between two successive solutions (calculated using the formula in the appendix B) is also shown, in Figure 4.

The results show a clear monotonic upper-bound convergence for all modes until an asymptotical value, confirming the theoretical result of upper-bound convergence of finite elements for modal analysis. Note that these asymptotical values are different for the python and Siemens NX models, which will be explained later. The relative change in between the fourth and fifth iteration are lower than respectively 0.1% and 0.01% for the NX and python models.

Therefore, after 5 iterations, the results of the python model have converged to their asymptotic values with relative changes lower than 0.1%.
From this number of elements per beam, the discretization errors are well below error tolerances and are considered negligible and the computational model is considered *verified*.

[1] Euler-Bernoulli theory assumes plane sections remain plane and **perpendicular** to the neutral axis, neglecting shear deformation. Timoshenko theory relaxes this perpendicularity constraint, accounting for shear.



(a) Mode 1

(b) Mode 2

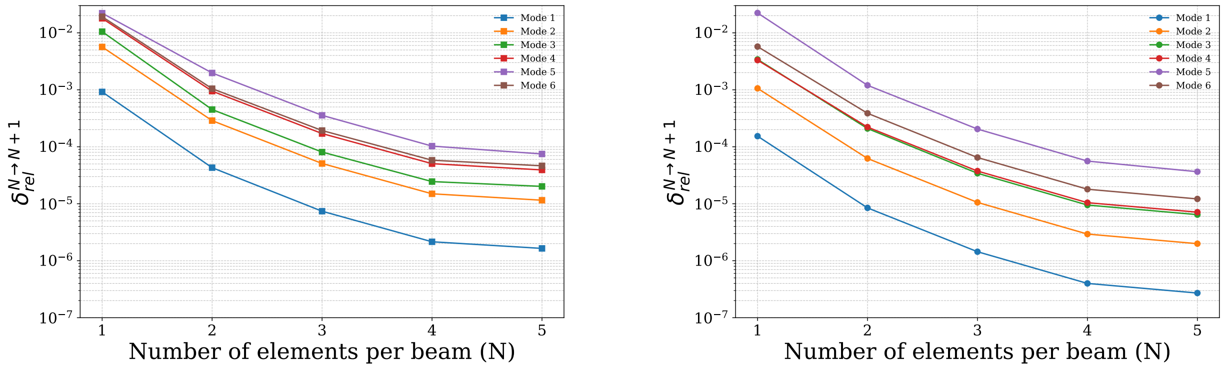
(c) Mode 3

(d) Mode 4

(e) Mode 5

(f) Mode 6

Figure 3: Evolution of the first six natural frequencies [Hz] of the models as a function of the number of finite elements per beam



(a) NX

(b) python

Figure 4: Relative improvements of natural frequencies using 1, 2, 3, 4, 5, and 10 elements per beam for results from Siemens NX and python.

1.2 Natural frequencies and modes

An analysis of the modal response of the two models with $N = 10$ elements per beam is performed to assess the *validity* of their solutions.

Figures 5 to 10 show the comparison of the python and Siemens NX models solutions. Table A shows the comparison of the natural frequencies associated to the first six modes between the two models.

Table A: Natural frequencies of the first six modes computed with $N = 10$ beam elements, using the models from Siemens NX and Python.

Model	**Mode 1**	**Mode 2**	**Mode 3**	**Mode 4**	**Mode 5**	**Mode 6**
Python [Hz]	2.573	6.844	10.856	12.916	13.866	14.877
NX [Hz]	2.569	6.827	10.816	12.864	13.856	14.816
Relative error [%]	0.14	0.25	0.36	0.40	0.07	0.41

Visually, figures 5 to 10 show all modes are identical between the python and NX models[2], except for the Mode 4 in which some transverse beams show important bending in python, while the NX model does not. This indicates that the python model is quite close to the reference one from the commercial software.

However, the natural frequencies show a slight difference $(0.1 - 0.4\%)$ in the value of the asymptotic values (see Figure 3 for convergence behavior and Table A for the exact asymptotic values), the python model having consistently higher values, meaning that this model is slightly stiffer.
This can be explained by the difference of theory used inside the two models : the python model uses Euler-Bernoulli beam theory, only accounting for bending and completely neglecting shear deformations. On the other hands, the model used in NX relies on the Timoshenko beam theory, which takes shear stress deformations into account (plane cross-sections no longer remain perpendicular to the deformed neutral axis), making the model more flexible and true to reality.

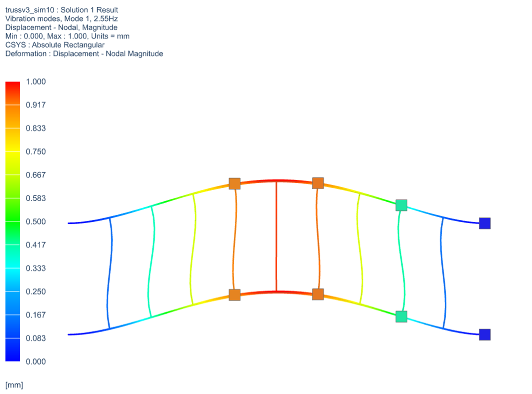
[2] A change of sign across the two models is irrelevant, as any mode is defined up to a multiplying constant

This difference between the two methods –induced by the shear deformations – is very small, and can be neglected for most applications, but is still interesting to notice.

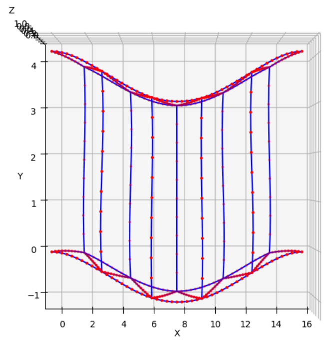
This difference is larger when the terms accounting for shear deformations in the stiffness matrix grow, i.e. when

$$\Phi_y = \frac{12EI_y}{L^2AG} \quad \text{and} \quad \Phi_z = \frac{12EI_z}{L^2AG} \not\ll 1. \tag{1.6}$$

It can be noted that the modes for which the relative errors are bigger are indeed the ones with the most pronounced bending.

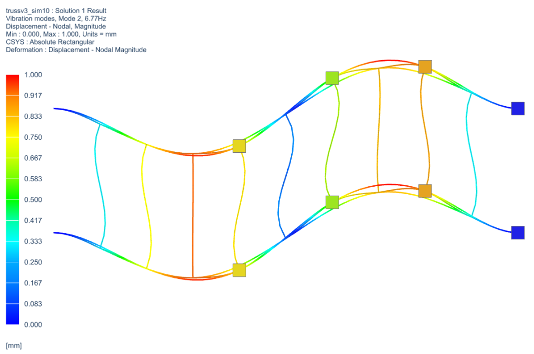


(a) XY view – Siemens NX

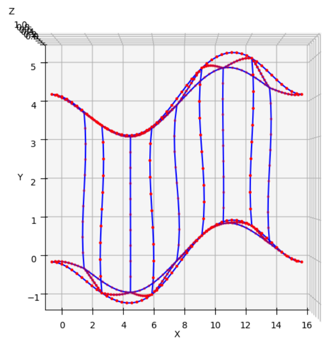


(b) XY view – Python

Figure 5: Mode 1 comparison (XY view) between NX and Python models, using $N = 10$ elements per beam. The XZ view is omitted due to negligible, not visible displacement.

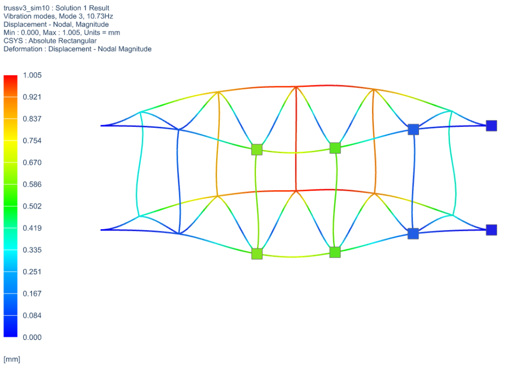


(a) XY view – Siemens NX

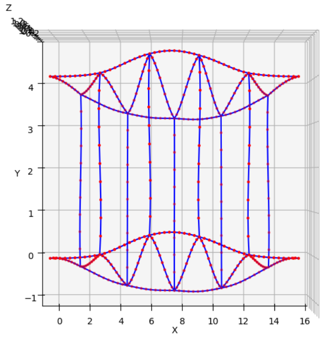


(b) XY view – Python

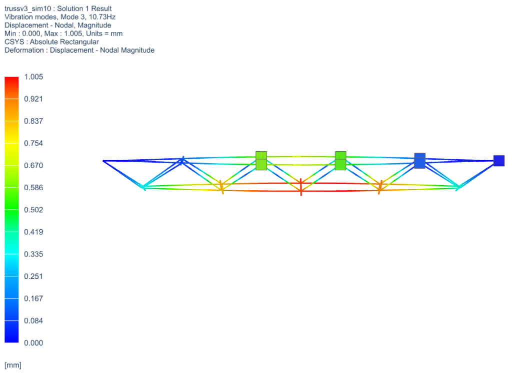
Figure 6: Mode 2 comparison (XY view) between NX and Python models, using $N = 10$ elements per beam. The XZ view is omitted due to negligible, not visible displacement.



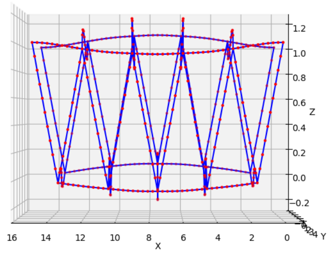
(a) XY view – Siemens NX



(b) XY view – Python

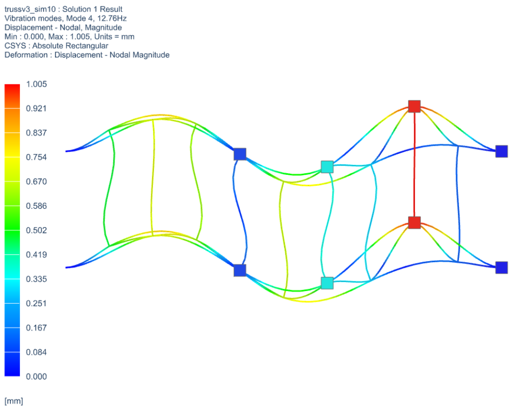


(c) XZ view – Siemens NX

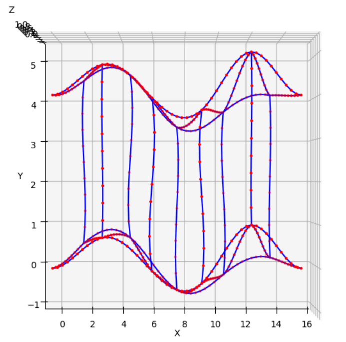


(d) XZ view – Python

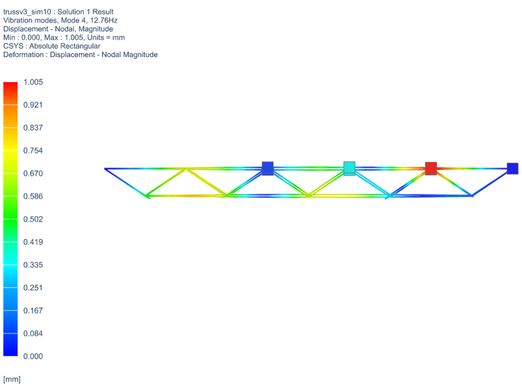
Figure 7: Mode 3 comparison between NX and Python models, using $N = 10$ elements per beam.



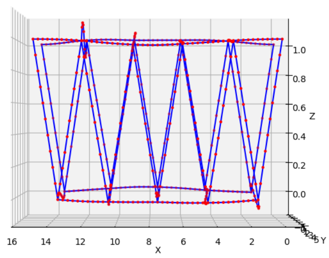
(a) XY view – Siemens NX



(b) XY view – Python

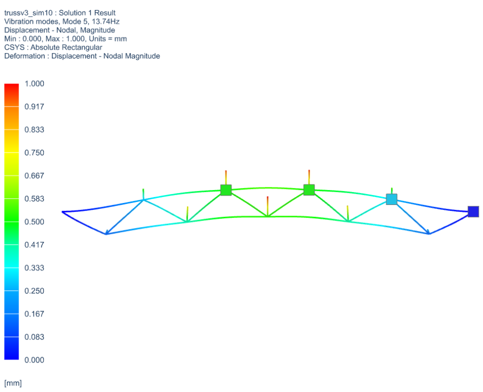


(c) XZ view – Siemens NX

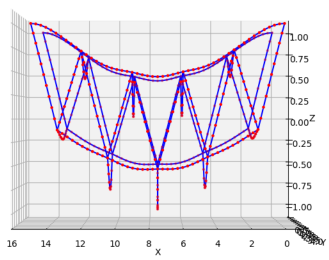


(d) XZ view – Python

Figure 8: Mode 4 comparison between NX and Python models, using $N = 10$ elements per beam.

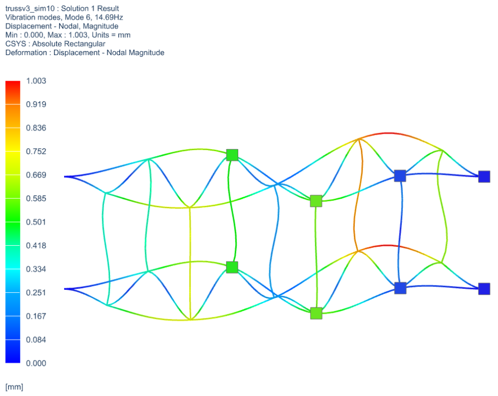


(a) XZ view – SiemensNX

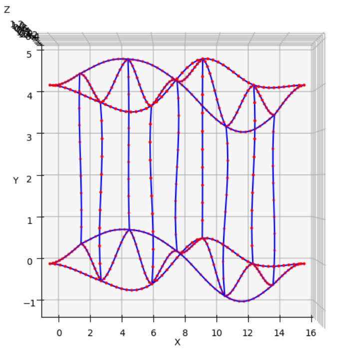


(b) XZ view – Python

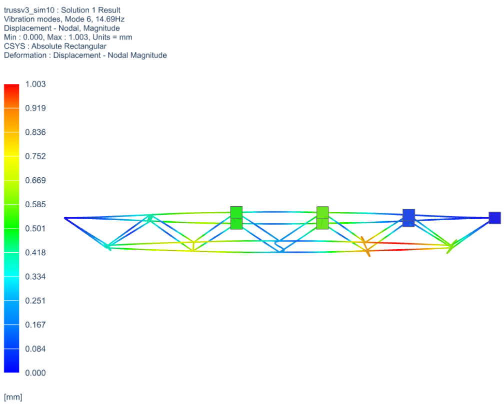
Figure 9: Mode 5 comparison (XZ view) between NX and Python models, using $N = 10$ elements per beam. The XY view is omitted due to negligible, not visible displacement.



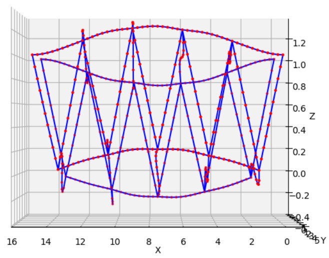
(a) XY view – Siemens NX



(b) XY view – Python



(c) XZ view – Siemens NX



(d) XZ view – Python

Figure 10: Mode 6 comparison between NX and Python models, using $N = 10$ elements per beam.

1.3 Total mass using rigid body mode in translation

To further validate the finite element model, the total mass is computed using a rigid body mode (RBM) in translation.
First, the real mass of the model is computed with

$$m_{tot} = m_{lumped} + \sum_e \rho A_e l_e \tag{1.7}$$

where A_e and l_e are the cross section area and the length of the beam e, and $m_{lumped} = 500$ kg is the lumped mass of the model.
This method yields a total mass of $m_{tot,analytical} = 1681.85$ kg.
To compute the mass of the model using its mass matrix, a RBM in translation can be used. A RBM is a displacement mode to which a zero strain energy is associated, i.e. $\mathbf{q}_{rbm}$ such that

$$\mathbf{K}_s \mathbf{q}_{rbm} = \mathbf{0}. \tag{1.8}$$

The mass of the structure associated to a mode is calculated by determining the generalized mass associated to it, and normalizing with regards to the mode :

$$m_{tot,FEM} = \frac{\mathbf{q}_{rbm}^T \mathbf{M}_s \mathbf{q}_{rbm}}{||\mathbf{q}_{rbm}||^2}. \tag{1.9}$$

Using the translation mode

$$\mathbf{q}_i = \{u_{ix}, u_{iy}, u_{iz}, \theta_{ix}, \theta_{iy}, \theta_{iz}\}^T = \{1, 1, 1, 0, 0, 0\}^T, \quad i \in \{1, N_{nodes}\} \tag{1.10}$$

yields a mass $m_{tot,FEM} = 1681.851$ kg too, confirming that the finite element model correctly conserves the total physical mass.
The mass computed by Siemens NX during the simulation is also $m_{NX} = 1681.851$ kg, the same as the python model, confirming that the difference in natural frequencies observed in Table A is due to a higher stiffness in the python model.

2 Stationary response

In this section, the canopy is submitted to a purely transverse load along the Y-axis, due to heavy machinery to repair damage. The force is applied at a single node, represented in green in Figure 11. The load is assumed to be a cosine wave, with an amplitude of 500 N and a frequency of 2.4 Hz, illustrated in Figure 12. For accuracy and computational efficiency, the number of elements per beam has been fixed to 5, according to the convergence study in the previous section.

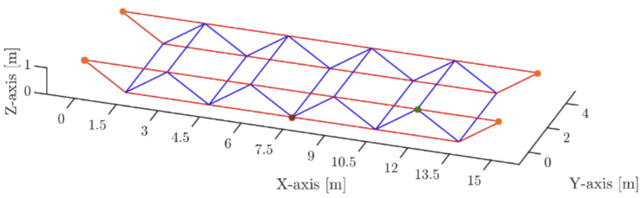


Figure 11: Model of the steel canopy used for the analysis of the stationary response.

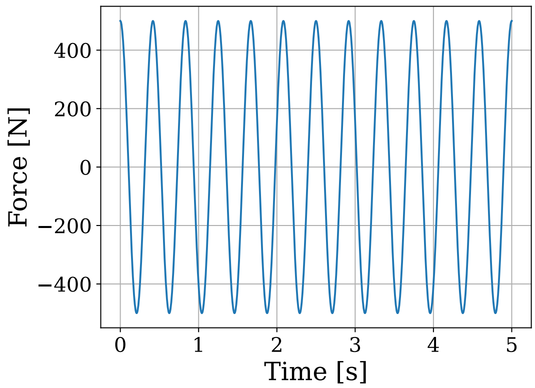


Figure 12: Time evolution of the purely transverse load, along Y-axis, at the excitation node.

2.1 Exact stationary response

In the case of a cosine wave force applied to an undamped structure, the equation of motion can be expressed as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathrm{s}\cos(\omega t)\ . \tag{2.1}$$

where s is the force vector, i.e. magnitude of force applied to each DOF. Hence, the exact solution of the problem is written

$$\mathbf{q} = \mathbf{x}\cos(\omega t), \tag{2.2}$$

where $\mathbf{x}$ is the spatial response, which can be computed through $\mathbf{x} = \mathbf{H}(\omega)\ \mathrm{s}$ with $\mathbf{H}(\omega) = (\mathbf{K} - \omega^2\ \mathrm{M})^{-1}$. The computation of the FRF matrix, $\mathbf{H}(\omega)$, will give the exact stationary response, depicted in Figure 13. This response will serve as the reference for the next analyses.

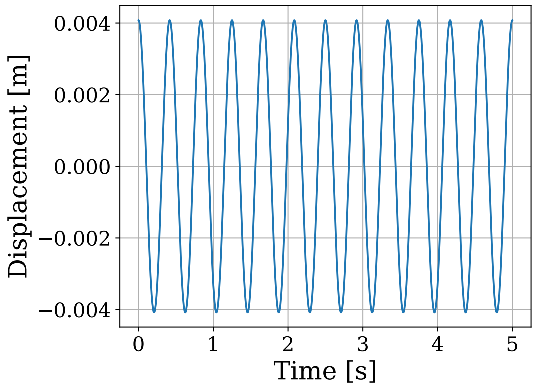


Figure 13: Time evolution (s) of the exact stationary response at the excitation node (m), computed with the FRF matrix.

The discrete Fourier transform (DFT) can be used to go from the time domain to the frequency domain. For a cosine wave, it will only take two values, one at the frequency of the applied force, and the other at the opposite of that frequency. In Figure 14, only the positive frequencies are represented because of the symmetry of the function, and the peak is effectively occurring at 2.4 Hz, the frequency of the cosine force.

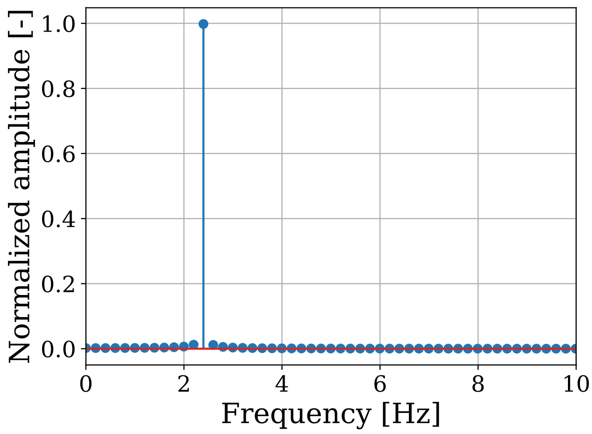


Figure 14: Normalized amplitude, computed with the discrete Fourier transform, with respect to the frequency [Hz].

2.2 Approximation methods

The main principle of the two approximation methods lies in computing the response of the system without taking into account all the modes of this system. Figure 15 shows a representation of the truncation that is done : the green part represents the frequency range in which modes are retained, the red part is the range in which the modes are not retained. The effect of this red part on the system is where the two methods differ.

Indeed, the mode displacement method neglects completely all the modes corresponding to the natural frequencies outside the frequency range of interest. It will simply compute the response with the first

k mode shapes. Whereas the mode acceleration method considers a quasi-static response for the red part, and only neglects inertia forces of these higher modes. This will lead to a compromise between computational cost and accuracy of the response, which will be explained later on. In the case of a harmonic excitation, with a cosine wave form, the responses can be obtained using both formulas,

$$\mathbf{x}_{dis} = \left(\sum_{s=1}^{k} \frac{\mathbf{x}_s\, \mathbf{x}_s^T}{(\omega_s^2 - \Omega^2)\, \mu_s} \right) \mathbf{s}\ , \tag{2.3}$$

$$\mathbf{x}_{acc} = \left(\mathbf{K}^{-1} + \Omega^2 \sum_{s=1}^{k} \frac{\mathbf{x}_s\, \mathbf{x}_s^T}{(\omega_s^2 - \Omega^2)\, \omega_s^2\, \mu_s} \right) \mathbf{s}\ , \tag{2.4}$$

where k is the number of mode shapes retained, lower or equal to the number of DOFs of the system, and $\mathbf{x}_s$ and μ_s are the eigenvector and generalized mass corresponding to the s^{th} natural frequency ω_s.

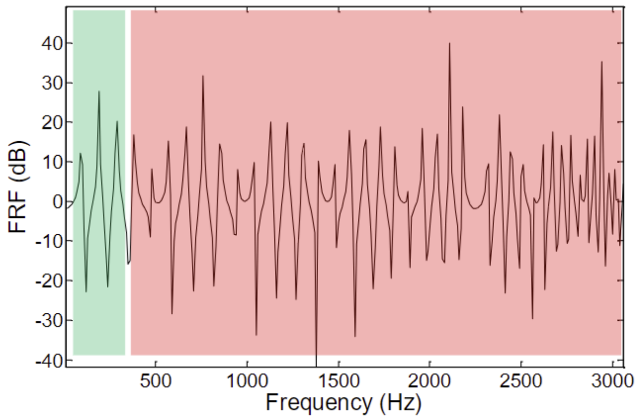


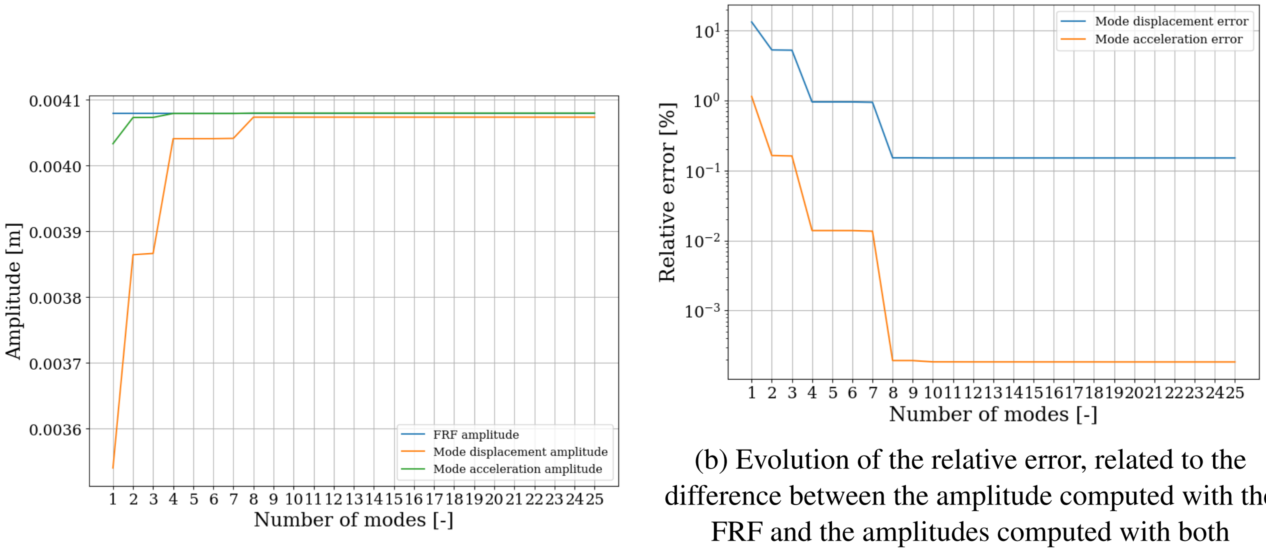
Figure 15: Representation of the frequency range of interest, in a frequency response diagram illustrating the amplitude [dB] with respect to the frequency [Hz]. Extract of lecture slides from L. Salles, *Theory of vibration*.

2.2.1 Convergence study

A convergence study is performed to assess the accuracy and rate of convergence of both methods. The results are shown in Figure 16 and Figure 16b.
The mode acceleration (MA) method shows rapid convergence achieving a relative error of 1% with only a single mode. In contrast, the mode displacement (MD) method's amplitude initial error is significant, leading to a higher number of modes required to reach similar 1% error.
This highlights the importance of considering the quasi-static response of the truncated high-frequency modes, rather than neglecting them entirely.

Furthermore, Figure 16b reveals that while both methods follow identical convergence trends due to the modal participation, they differ in magnitude : there is an initial vertical shift due to the quasi-static behavior correction in the MA method and a faster rate of convergence (steeper jumps) for the MA method; because its residual error come from neglected inertia forces (scaling as $1/\omega_s^4$), whereas the MD method's error come from neglected displacements (scaling as $1/\omega_s^2$).

For both figures, a maximum arbitrary number of modes has been fixed in a first approach. However, Figure 16 illustrates that this number can be dramatically reduced since a plateau occurs



(a) Evolution of the amplitude of the response at the excitation node [m], for both methods, varying with respect to the number of modes taken into account.

(b) Evolution of the relative error, related to the difference between the amplitude computed with the FRF and the amplitudes computed with both methods, of the response at the excitation node [%], for both methods, varying with respect to the number of modes taken into account.

Figure 16: Convergence of the amplitude of the response for both methods regarding the number of modes used.

after approximately 10 modes. As a result, only the first 10 modes will be considered for next analyses. Furthermore, other smaller plateaus are observed in the graphs. Indeed, between mode 2 and 3, and between mode 4 and 7, the amplitude, or even the relative error, remain constant. Figure 17 helps to understand the contribution of each mode on the response. The low contributions of mode 3 and mode 5 stated previously are verified by being local minima of each curve. In addition, the contribution of each mode tends to drop as the number of modes increases.

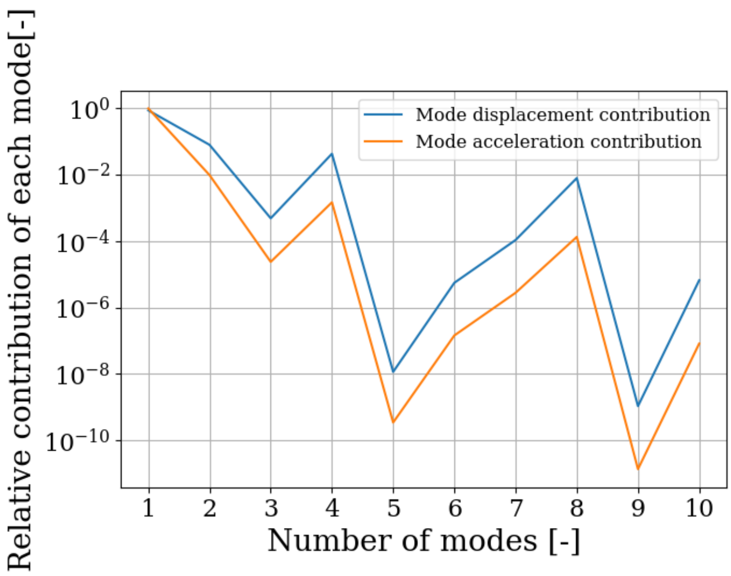


Figure 17: Representation of the extra value added by the i^{th} mode to the amplitude computed with the $i-1^{th}$ mode.

This trend can be explained through modal superposition. The total displacement response $\mathbf{q}(t)$

is expressed as a linear combination of the n mass-normalized eigenmodes $\mathbf{x}_s$ and the generalized normal coordinates $\eta_s(t)$:

$$\mathbf{q}(t) = \sum_{s=1}^{n} \mathbf{x}_s \eta_s(t) \; . \tag{2.5}$$

When projecting the motion equations on the modal basis and considering orthogonality equation, the system is composed of n 1D uncoupled equations. The motion of the s^{th} coordinate is given by

$$\ddot{\eta}_s(t) + \omega_s^2 \eta_s(t) = \phi_s(t) \; , \tag{2.6}$$

Where $\phi_s(t) = \frac{\mathbf{x}_{(s)}^T \; \mathbf{p}(t)}{\mu_s}$ is the modal force factor. By separating $\mathbf{p}(t)$ into a spatial distribution vector $\mathbf{g}$ and a time function $\varphi(t)$, the solution for the normal coordinate (i.e. **the contribution of the** s^{th} **mode to the structural displacement**) is given by

$$\eta_s(t) = \underbrace{\frac{\mathbf{x}_s^T \mathbf{g}}{\mu_s}}_{\text{Spatial term } X_s} \; \underbrace{\frac{1}{\omega_s} \int_0^t \sin(\omega_s(t-\tau))\varphi(\tau) d\tau}_{\text{Temporal term } \theta_s(t)} \; , \tag{2.7}$$

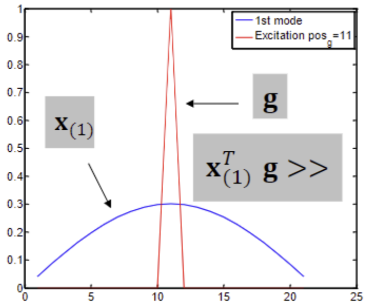
highlighting the two distinct factors driving the convergence behavior observed in Figure 17:

- The Spatial Factor X_s and
- The Temporal Factor $\theta_s(t)$.

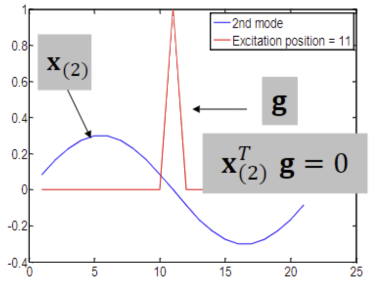
The spatial factor, $X_s = \frac{\mathbf{x}_{(s)}^T \mathbf{g}}{\mu_s}$, is governed by the orthogonality relationship between the mode shape and the load vector. As a consequence, if a mode shape is orthogonal to the load vector, *i.e.* $\mathbf{x}_{(s)}^T \mathbf{g} = 0$, then it will not contribute to the response. In the same way, a low product will lead to a negligible contribution. Visually, it is straightforward to understand this relationship with Figure 18.Given that the load is applied along the Y-axis at the green node (Figure 11), mode shapes with large Y-axis deformation at that specific node will have a significant contribution. For instance, Figure 17 clearly illustrates that modes 2 and 4 are important contributors to the response, while they correspond to modes with large Y-axis deformation at the excitation node (Figures 6 and 8). In contrast, modes 3 and 5 are low contributors (Figure 17), which aligns with the Figures 7 and 9. Overall, the plateaus depicted in Figure 16 are a direct result of these orthogonal relationships, because it leads in the additions of a quasi zero value.

The temporal factor, $\theta_s(t)$, is inversely proportional to ω_s, according to Equation 2.7. Consequently, as the natural frequency increases, which corresponds to modes related to higher natural frequencies, the temporal factor decreases. This explains why in Figure 17, the modal contribution decreases for higher modes.

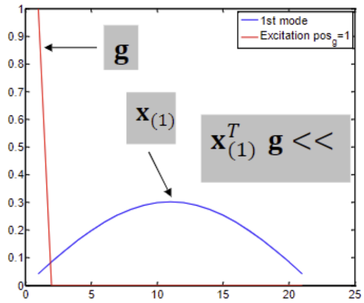
The overall modal contribution is governed by two factors : the orthogonality relationship between the modes and the load vector, but also the natural frequencies related to the modes. As a consequence, modes with a negligible contribution can be truncated from the computation of the response. This reduces the number of terms to be evaluated, and even increases the rate of convergence.



(a) High contribution, $\mathbf{x}_{(s)}^T\mathbf{g} \gg$



(b) No contribution, $\mathbf{x}_{(s)}^T\mathbf{g} = 0$



(c) Low contribution, $\mathbf{x}_{(s)}^T\mathbf{g} \ll$

Figure 18: Representation of the orthogonality relationship between mode shape and load vector in three different cases. Extract from L. Salles, *Theory of vibration*.

2.2.2 Approximated time responses

As deduced from the previous study, a coherent number of modes to consider sits around 10 modes. To illustrate the time response for each method, $k = 10$ will be fixed, because there is no need to compute the time response for multiple k, otherwise it will lead to the same conclusion as already discussed before.

As depicted in Figure 19, the curves for both approximation methods and the exact solution are coincident, which is due to the choice of k that induces a relative error of less than $10^{-3}\%$ (Figure 16b).

2.3 PSD and RMS value of the acceleration of the response

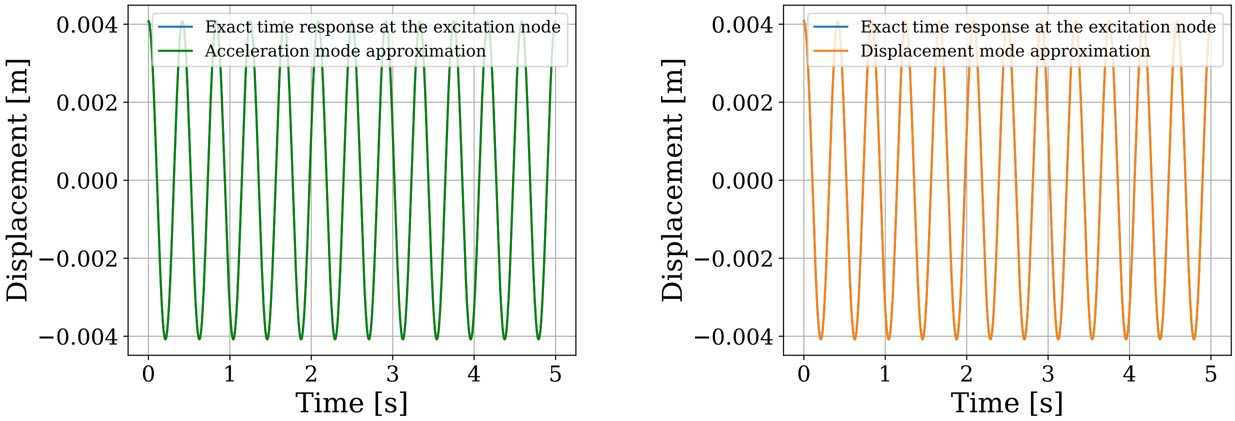
The power spectral density (PSD) of a general cosine signal can be defined as

$$G_f(\omega) = \frac{1}{2}\, A^2\, \delta(\omega - \Omega) \tag{2.8}$$

with $\delta(\omega - \Omega)$ the Dirac function. Given that

$$\int \delta(\omega)\, d\omega = 1 \tag{2.9}$$

by definition of the Dirac function, the integral of the PSD over the frequency should be equal to $\frac{1}{2}\, A^2$. To compute the PSD, the only non-zero value of the function must be $\frac{1}{2}\, A^2\, \frac{1}{d\omega}$, resulting in a cancellation of the infinitesimal step and the verification of the desired result.



(a) Mode acceleration method.

(b) Mode displacement method.

Figure 19: Time evolution of the displacement [m] for both methods, with $k = 10$ modes retained, i.e. when a plateau is reached .

In this case, the signal corresponds to the acceleration of the response, which can be determined by pre-multiplying the amplitude vector $\mathbf{x}$ by $-\Omega^2$, the excitation frequency. In other words,

$$A = | - \Omega^2 \, \mathbf{x}| \, . \tag{2.10}$$

This leads to the evaluation of the RMS value

$$\sqrt{\overline{f^2}} = \frac{A}{\sqrt{2}} = 0.6560 \text{ [m/s}^2\text{]} \, , \tag{2.11}$$

and the representation of the PSD over the frequency (Figure 20). Furthermore, still by definition of the Dirac function, it gives $\delta(\omega - \Omega) = 1 \quad \text{if} \quad \omega - \Omega = 0$. Consequently, the non-zero value of the PSD occurs when $\omega = \Omega \simeq 15$ rad/s, which is verified in Figure 20.

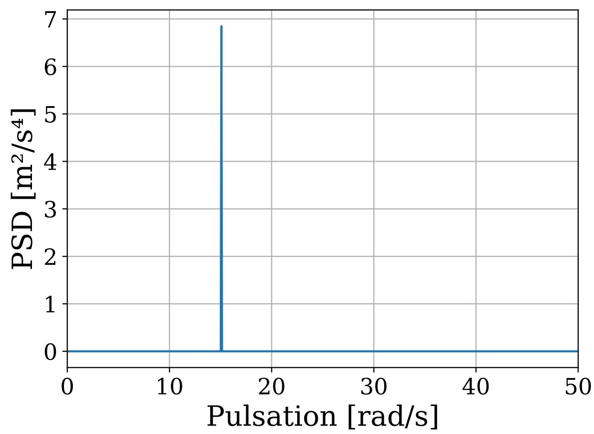


Figure 20: Power spectral density of the acceleration of the response [m^2/s^4]with respect to the pulsation [rad/s].

To check the validity of the results, as previously said, the integral of the PSD over the frequency should be equal to the RMS value squared,

$$\int G_f(\omega)\, d\omega = 0.4303 \text{ [m}^2\text{/s}^4\text{]} \ . \tag{2.12}$$

2.4 Comfort level

Concerning the comfort level, based on Figure 21, the maximum time of exposure on the canopy at the excitation node is roughly 2.5 hours . This point (blue point in Figure 21), is obtained thanks to the excitation frequency, $\Omega = 2.4$ Hz, the RMS value, $\sqrt{\overline{f^2}} = 0.6560$ m/s^2, and a lateral acceleration in this case.

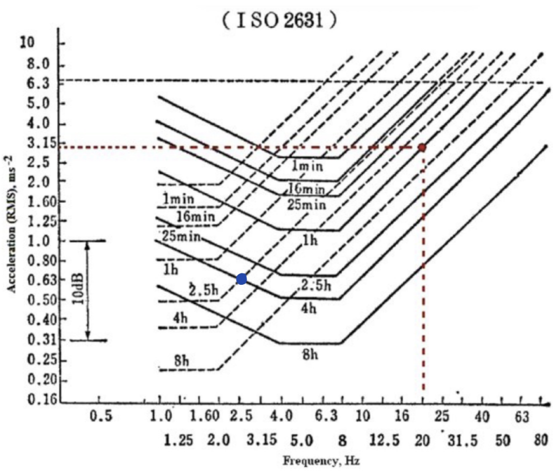


Figure 21: ISO 2631 curves defining equal fatigue-decreased proficiency boundaries for a vertical acceleration (solid line) and a lateral acceleration (dashed line).

3 Transient response

In this section, the damping of the structure will be considered : a modal damping ratio of 1% for the first two modes. The objective is to obtain the exact transient response of the structure due to an impact at the brown node in Figure 11, corresponding to a force of 7000 N for a duration of 0.01 s, still purely transverse along the Y-axis.

3.1 Damping matrix

The general equation of motion can thus be rewritten, considering viscous damping here, as

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{p}(t) \ , \tag{3.1}$$

where $\mathbf{C}$ is the damping matrix.
To decouple the equations of motions using the mode shapes from the undamped system, two assumptions are made :

- Weak damping assumption : damping introduces a first order correction on the natural frequencies of the undamped system : $\lambda_k \cong i\omega_{0k} - \frac{\beta_{kk}}{2\mu_k}$ and a negligible correction of the mode shapes : $\mathbf{z}_{(k)} \cong \mathbf{x}_{(k)}$.
- modal assumption is made : the damping matrix is a diagonal in the modal basis (equal to the undamped modal basis by the weak damping assumption), *i.e.* $\beta_{rs} = \mathbf{x}_{(r)}^T \, \mathbf{C} \, \mathbf{x}_{(s)} = 0$ for $r \neq s$.

One way to satisfy this condition, the one chosen here, is to consider proportional (Rayleigh) damping, written as

$$\mathbf{C} = \alpha \, \mathbf{K} + \beta \, \mathbf{M} \quad . \tag{3.2}$$

which satisties modal orthogonality by the orthogonality of the mass and stiffness matrices themselves.
Coefficient α and β are determined by forming $\mathbf{x}_{(r)}^T \mathbf{C} \mathbf{x}_{(r)}$ leading to :

$$\varepsilon_r = \frac{1}{2} \, (\alpha\omega_{0r} + \frac{\beta}{\omega_{0r}}) \quad , \tag{3.3}$$

with ε_r the damping ratio of mode r,. This is equivalent to solving the system :

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1/\omega_{01} & \omega_{01} \\ 1/\omega_{02} & \omega_{02} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad . \tag{3.4}$$

The damping ratios and the natural frequencies are determined by modal testing and, in this case, the ratios are both set to 1% on the first two modes. So, the solutions are $\alpha = 3.38 \ 10^{-4}$ s and $\beta = 2.35 \ 10^{-1}$ s^{-1}. From Equation 3.2, the damping matrix can now be computed.

To evaluate the first six damping ratios, two formulas can be used : one is simply the Equation

3.3, and the other is by using the uncoupled normal equations system, which leads to the equation :

$$\varepsilon_r = \frac{\beta_r}{2\omega_r \mu_r} \tag{3.5}$$

with μ_r the generalized mass corresponding to the rth natural frequency, ω_r. Table B depicts the different damping ratios. One can easily verify that all the values belong to the interval ≈[1% - 10%], which corresponds to experimental values from the civil engineering structures domain.

Table B: Damping ratios corresponding to the first six modes

Mode [-]	Damping ratio [-]
1	0.010
2	0.010
3	0.013
4	0.015
5	0.016
6	0.017

3.2 Time integration

Direct time integration performs the computation of the exact solution of a system, and that for general systems and excitations, without the need of performing a modal expansion. The Newmark algorithm will be used as the time integration method. It is based on the general equation of motion for a damped system :

$$\mathbf{M}\ddot{\mathbf{q}}_{n+1} + \mathbf{C}\dot{\mathbf{q}}_{n+1} + \mathbf{K}\mathbf{q}_{n+1} = \mathbf{p}_{n+1} \quad , \tag{3.6}$$

with $h = t_{n+1} - t_n$. The three unknowns $\ddot{\mathbf{q}}_{n+1}$, $\dot{\mathbf{q}}_{n+1}$, $\mathbf{q}_{n+1}$ will be evaluated thanks to the approximation by their Taylor series, and by combining them with the introduction of two coefficients, $\gamma \in [0, 1]$ and $\beta \in [0, 1/2]$, which control the explicit-implicit repartition of the schemes. The Newmark formulas are therefore given by :

$$\begin{aligned} \dot{\mathbf{q}}_{n+1} &= \dot{\mathbf{q}}_n + (1-\gamma)h\ddot{\mathbf{q}}_n + \gamma h \ddot{\mathbf{q}}_{n+1} + \mathbf{r}_n \\ \mathbf{q}_{n+1} &= \mathbf{q}_n + h\dot{\mathbf{q}}_n + \left(\frac{1}{2} - \beta\right) h^2 \ddot{\mathbf{q}}_n + \beta h^2 \ddot{\mathbf{q}}_{n+1} + \mathbf{r}'_n \end{aligned} \tag{3.7}$$

with $\mathbf{r}_n$ and $\mathbf{r}'_n$ the errors. Injecting those two equations into Equation 3.6 for a linear system, a formula for $\ddot{\mathbf{q}}_{n+1}$ is deduced, and the Newmark algorithm following these developments is shown in Figure 22.

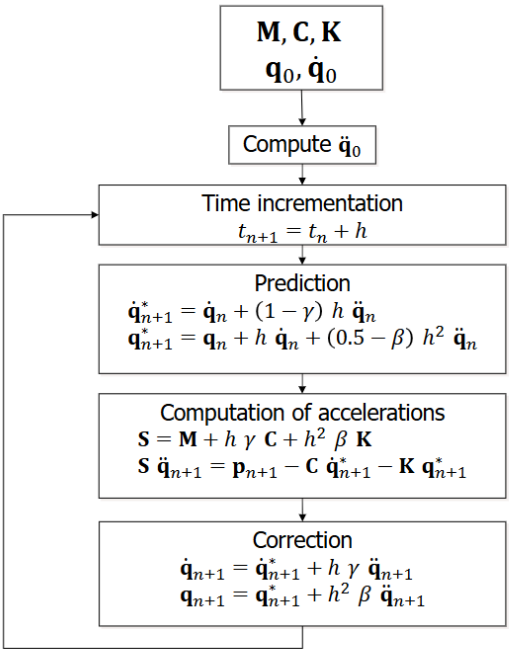


Figure 22: Flowchart of the Newmark Algorithm for linear systems. Extract from L. Salles, *Theory of vibration*.

3.2.1 Integration parameters

Numerical stability

The numerical stability of the Newmark integration is determined by the integration parameters γ and β. Unconditional stability is guaranteed if

$$\begin{aligned} \gamma &\geq \frac{1}{2}\,; \\ \beta &\geq \frac{1}{4}\left(\gamma + \frac{1}{2}\right)^2. \end{aligned} \tag{3.8}$$

Figure 23 shows these stability boundaries. When $\gamma < 1/2$, the scheme becomes unstable; for pairs of parameter values in which γ satisfies its condition but not β, it is required to use a time step h smaller than a critical value related to the highest natural frequency of the discrete system.

To illustrate this, the transient response of the truss is computed for two distinct parameter sets with fixed time of integration T and time step h (Figure 24).
In Figure 24a, the time response follows a decreasing trend, characteristic of the physical damping of structure. The chosen value of γ ensures to be in the stable region, which is confirmed by the bounded solution. However, in Figure 24b, the displacement takes increasingly large values as time increases, which is the complete opposite of a damped structure. This behavior is associated to the unstable region, due to $\gamma < \frac{1}{2}$. This highlights the importance of choosing integration parameters satisfying Equation 3.8 to obtain a feasible solution at the excitation node.

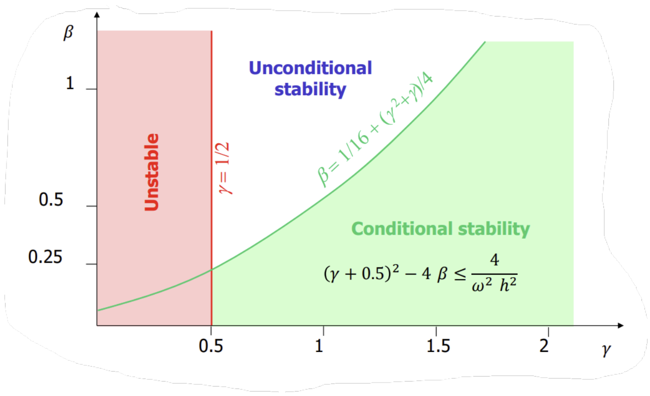
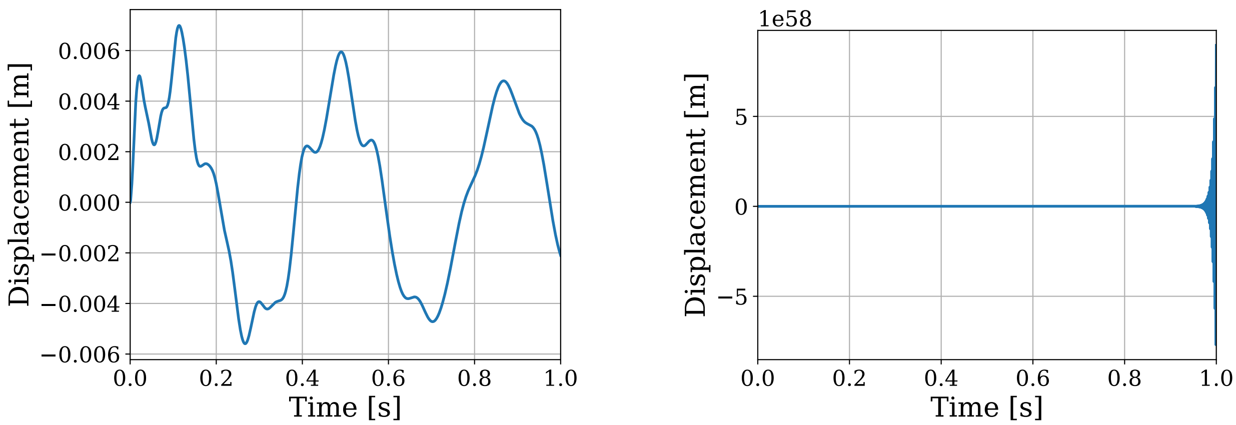


Figure 23: Stability regions of the solution depending on the parameters γ and β. Extract from L. Salles, *Theory of vibration*.



(a) Stable solution, $\gamma = \frac{3}{4}$, $\beta = 0.429$.

(b) Unstable solution, $\gamma < \frac{2}{5}$, $\beta = 0.222$.

Figure 24: Time response at the excitation node in the unstable and stable cases, with $T = 1$[s] and $h = 0.001$[s].

Numerical damping and accuracy

In addition to stability, values of the integration introduce numerical damping $\bar{\varepsilon}$ and periodicity error $\frac{\Delta T}{T}$ (*i.e.* a frequency shift, contributing to the accuracy of the solution. These errors are given by

$$\begin{aligned}\bar{\varepsilon} &= \frac{1}{2}\left(\gamma - \frac{1}{2}\right)\omega h - \frac{1}{2}\left(\beta + \frac{\gamma}{2} - \frac{1}{4}\right)\left(\gamma - \frac{1}{2}\right)\omega^3 h^3 + \mathcal{O}(\omega^5 h^5)\\ \frac{\Delta T}{T} &= \frac{1}{2}\left(\beta + \frac{11}{48} + \frac{\gamma}{4}(\gamma - 3)\right)\omega^2 h^2 + \mathcal{O}(\omega^4 h^4)\end{aligned} \tag{3.9}$$

The different existing integration schemes for the Newmark algorithm are shown in Figure 25. Among them, only the two lasts are unconditionally unstable. Therefore all other schemes should not be retained as they can lead to diverging, non energy-conservative results. Thus, the two schemes studied are :

- The average constant acceleration, with parameters $\gamma = 0.5$ and $\beta = 0.25.$;
- The average constant acceleration modified with parameters $\gamma = 0.5 + \alpha = 0.9142$ and $\beta = \frac{(1+\alpha)^2}{4} = 0.5$.

			Stability limit	Accuracy	
				Amplitude error	Periodicity error
Algorithm	γ	β	$\omega\ h$	$\rho - 1$	$\frac{\Delta T}{T}$
Purely explicit	0	0	0	$\frac{\omega^2\ h^2}{4}$	$-$
Central difference	$\frac{1}{2}$	0	2	0	$-\frac{\omega^2\ h^2}{24}$
Fox & Goodwin	$\frac{1}{2}$	$\frac{1}{12}$	2.45	0	$O(h^3)$
Linear acceleration	$\frac{1}{2}$	$\frac{1}{6}$	3.46	0	$\frac{\omega^2\ h^2}{24}$
Average constant acceleration	$\frac{1}{2}$	$\frac{1}{4}$	∞	0	$\frac{\omega^2\ h^2}{12}$
Average constant acceleration (modified)	$\frac{1}{2}+\alpha$	$\frac{(1+\alpha)^2}{4}$	∞	$-\alpha\ \frac{\omega^2\ h^2}{2}$	$\frac{\omega^2\ h^2}{12}$

Figure 25: Different integration schemes for the Newmark algorithm

By replacing these values in Equation 3.9, it can be computed that the first scheme introduces zero numerical damping ($\bar{\varepsilon} = 0$), which in the case of an undamped system, may retain spurious oscillations caused by the FE discretization. For damped systems, these spurious high-frequency modes are physically damped so there is no concern related to this.
The second scheme, however, by setting $\gamma = 0.91 > 0.5$, introduces numerical damping given by

$$\bar{\varepsilon}_\alpha = -\frac{\alpha}{2}\omega^2 h^2. \tag{3.10}$$

Furthermore, both schemes exhibit a periodicity error equal to

$$\frac{\Delta T}{T} = \frac{\omega^2 h^2}{12} \tag{3.11}$$

The time responses corresponding to these schemes are depicted in Figure 26. Visually, in the second case the response takes lower values and is much smoother, due to the numerical damping. In contrast, their periods do not differ, as the both show the same periodicity error.

Conclusion

In conclusion, the average constant acceleration scheme ($\gamma = 1/2$ and $\beta = 1/4$) should be retained, as it introduces only a second-order error on periodicity and no numerical damping, ensuring that the observed damping only comes from the structural damping matrix $\mathbf{C}$.

3.2.2 Time step

To decouple and quantify the two error sources identified in Equation 3.9—periodicity error and numerical damping—two distinct convergence studies are performed with respect to the time step h. The first one focuses on showing the periodicity error, which is an integration error that can not be eliminated from any integration scheme. For this first study, the average constant acceleration scheme ($\gamma = 1/2$ and $\beta = 1/4$) is used, so as to isolate the periodicity error.

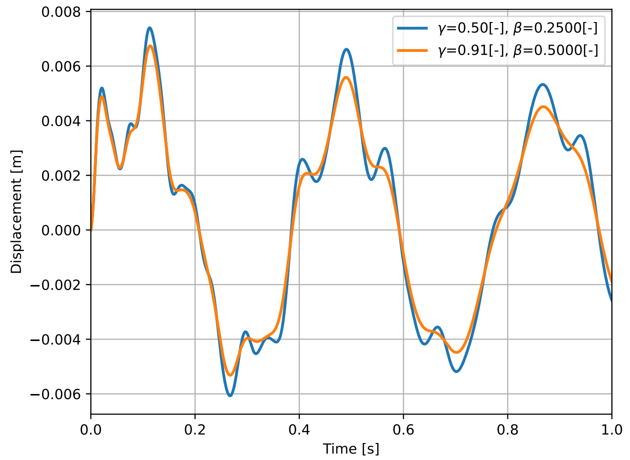


Figure 26: Time evolution of the response at the excitation node along the Y-direction with different integration parameters, and with $T = 1$[s] and $h = 0.001$[s].

The second one focuses on the numerical damping, which is not present in the average constant acceleration (the error is $\mathcal{O}(\omega^5 h^5)$, so the modified average constant acceleration scheme has to be used, so for this second study the parameters chosen are $\gamma = 0.91$ and $\beta = 0.5$.

There exists however one constraint on the time step, being that the force duration ($t_{force} = 0.01$ s) must be a multiple of the time step, i.e. $N_{step}\ h = t_{force}$ where $N_{step} \in \mathbb{Z}^+$. Otherwise, the integration would not be able to capture the effect of the applied force on the entirety of its duration. This directly imposes that the largest time step to be used is the force duration 0.01 s.

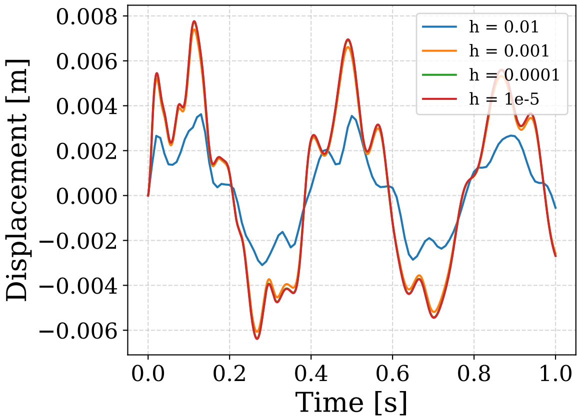
Periodicity error

Figure 27 and Figure 28 show the results of the convergence study on the periodicity error.
Amplitude of the responses should not be looked at since only the periodicity error is of interest in this part.

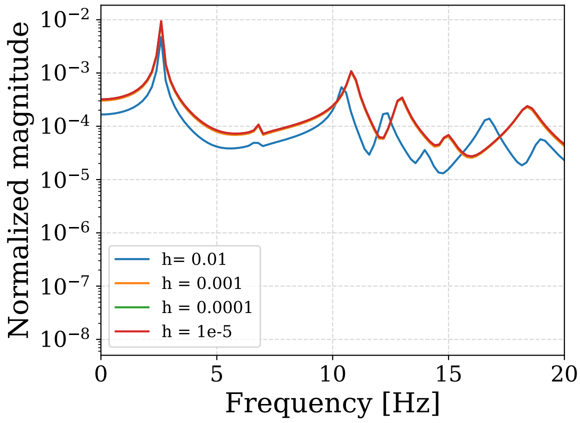
The time response in Figure 27a reveals only a minor phase lag for the largest time step ($h = 0.01$ s). However, the frequency domain analysis in Figure 27b provides greater insight: a distinct frequency shift growing as the frequency increases is observed for $h = 0.01$ s, which is symptomatic of the periodicity error. For time steps $h \leq 10^{-3}$ s, this error becomes negligible. This is confirmed quantitatively by Figure 28, which demonstrates the expected second-order convergence rate ($\mathcal{O}(h^2)$) of the average constant acceleration scheme. Furthermore, an analytical verification using Equation 3.9 yields a relative period error of $\frac{\Delta T}{T} \approx 10^{-5}$ for the first natural frequency. While this error remains low for the fundamental mode, it is noted that it scales with ω^2, implying significant phase shift for higher-frequency modes if the time step is not sufficiently refined.

Numerical damping

Figure 29 shows the results of the convergence study on the numerical damping. The amplitude error evolution is quantified in Figure 30.



(a) Time evolution of the response



(b) Normalized DFT of the response

Figure 27: Assessment of periodicity error with time step variation. (a) Early transient time response at the excitation node (0-1 s). (b) Normalized Discrete Fourier Transform computed over $T = 5$ s. Integration parameters: $\gamma = 0.5, \beta = 0.25$.

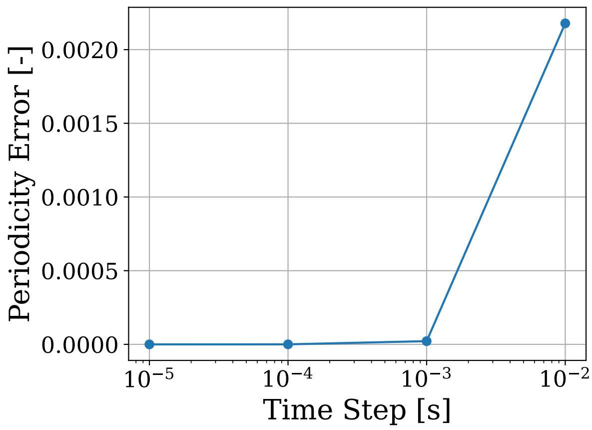
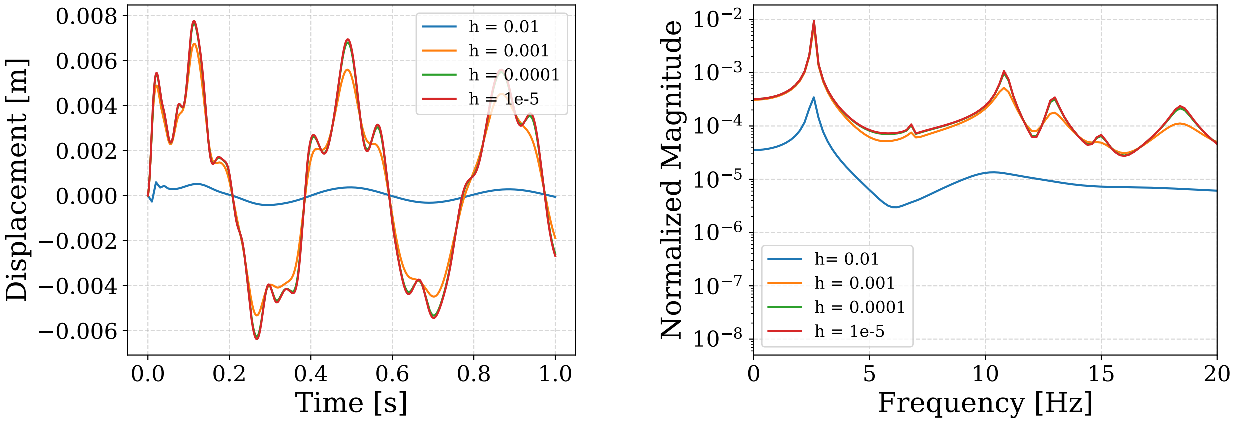


Figure 28: Evaluation of the periodicity error by varying the time step, considering the first natural frequency. Integration parameters: $\gamma = 0.5, \beta = 0.25$.

For the largest time step ($h = 0.01$ s), the system shows a significant damping. The peak amplitude is attenuated by approximately an order of magnitude compared to the converged solution. More notably, the frequency content in Figure 29b reveals that the numerical integration acts as a low-pass filter: modes 2 and above completely vanish after only 0.2 s, and are flattened and lowered on the frequency spectrum. This confirms that large time steps introduce excessive numerical damping. Refining the time step reduces this numerical dissipation. For $h = 10^{-3}$ s, the first mode is well-preserved, yet higher-frequency peaks remain attenuated. This observation aligns with the theoretical error term $\bar{\varepsilon} \propto \omega h$. Consequently, high-frequency modes are filtered out more aggressively than the first ones.

Finally, for $h \leq 10^{-4}$ s and smaller, the numerical damping becomes negligible, and the response converges to the true physically damped solution. Figure 30 confirms this convergence. Unlike the periodicity error (which scaled as $\mathcal{O}(h^2)$), the amplitude error in this damped scheme typically follows



(a) Time evolution of the response

(b) Normalized DFT of the response

Figure 29: Assessment of numerical damping with time step variation. (a) Early transient time response at the excitation node (0-1 s). (b) Normalized Discrete Fourier Transform computed over $T = 5$ s. Integration parameters: $\gamma = 0.91, \beta = 0.5$.

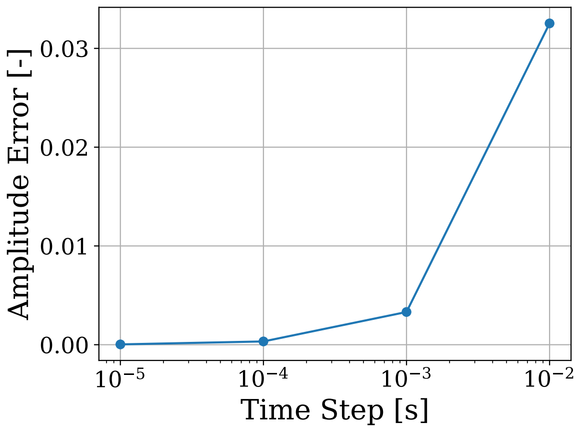


Figure 30: Evaluation of the amplitude error by varying the time step, considering the first natural frequency. Integration parameters: $\gamma = 0.91, \beta = 0.5$.

a first-order convergence ($\mathcal{O}(h)$), as shown in Equation 3.9, requiring finer time steps to achieve converged results and a good accuracy.

Conclusion

The convergence analysis reveals that the limiting factor for the time step depends on the integration scheme.

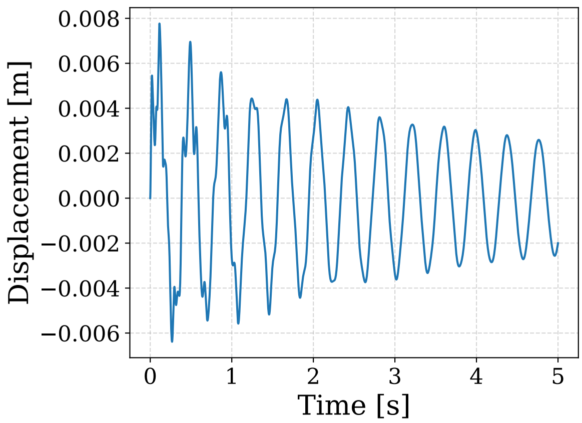
For dissipative schemes ($\gamma > 0.5$), the accuracy is limited by numerical damping. Since this amplitude error typically converges linearly ($O(h)$), a fine time step of $h = 10^{-4}$ s is required to prevent numerical damping.

For the average constant acceleration scheme ($\gamma = 0.5$), the accuracy is limited by the periodicity error. As it scales quadratically ($O(h^2)$), the constraint can be relaxed, and a time step of $h = 10^{-3}$ s yields sufficiently converged and accurate results.

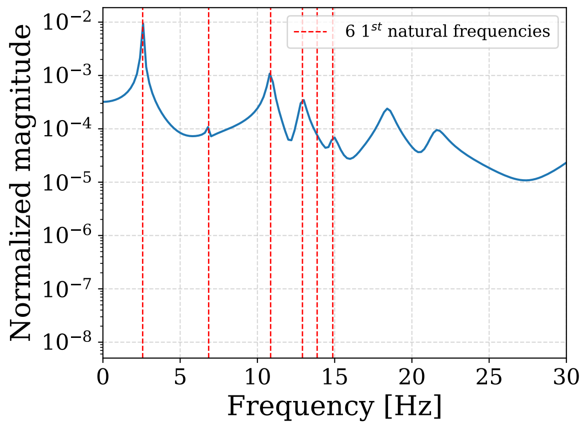
However, to minimize errors due to integration schemes in the remainder of the report, the conservative value of $h = 10^{-4}$ s will be used.

3.3 Time response and DFT of the response

Finally, the time response and the associated discrete Fourier transform obtained with the average constant acceleration scheme and a time step $h = 10^{-4}$ s are shown in Figure 37.



(a) Time evolution of the response



(b) Discrete Fourier transform of the response

Figure 31: Evolution of the response at the excitation node along the Y-direction with $T = 5$ s , $h = 10^{-4}$ s , $\gamma = 0.5$, $\beta = 0.25$.

4 Reduction methods

The objective of this section is to explore how to compute the behior of specific nodes of interest more efficiently than by solving the full system, while remaining accurate. To do so, only the three translational DOFs corresponding to the purple nodes in Figure 32 will be retained. Two techniques are employed and later compared with the full model: the Guyan–Irons *static condensation* and the Craig–Bampton *super-element* method. The comparison is based on modal analysis and time-domain response. The further analyses will be based on a full system built with $N = 10$ elements per beam.

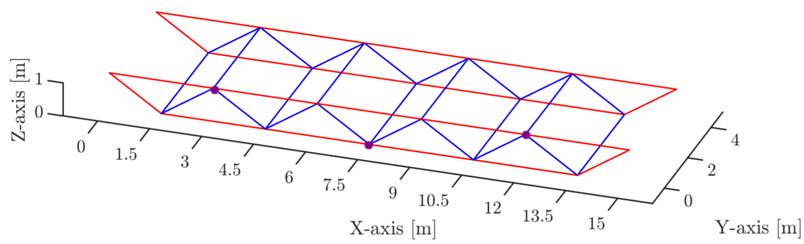


Figure 32: Model with the 3 retained nodes for the system reduction highlighted in purple.

Reduction is formulated by starting from the full eigenvalue problem

$$\mathbf{K}\mathbf{x} = \omega^2 \mathbf{M}\mathbf{x}, \tag{4.1}$$

and partitioning the displacement vector $\mathbf{x}$ into *retained* (R) and *condensed* (C) subsets,

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_R \\ \mathbf{x}_C \end{bmatrix}, \tag{4.2}$$

and partitioning matrices accordingly, resulting in the partitioned system of equations of motion

$$\begin{bmatrix} \mathbf{K}_{RR} & \mathbf{K}_{RC} \\ \mathbf{K}_{CR} & \mathbf{K}_{CC} \end{bmatrix} \begin{bmatrix} \mathbf{x}_R \\ \mathbf{x}_C \end{bmatrix} = \omega^2 \begin{bmatrix} \mathbf{M}_{RR} & \mathbf{M}_{RC} \\ \mathbf{M}_{CR} & \mathbf{M}_{CC} \end{bmatrix} \begin{bmatrix} \mathbf{x}_R \\ \mathbf{x}_C \end{bmatrix}. \tag{4.3}$$

Each reduction method consists in building its transformation matrix $\mathbf{R}$ given by

$$\mathbf{x} = \mathbf{R}\mathbf{x}_R, \tag{4.4}$$

leading to reduced matrices

$$\begin{aligned} \tilde{\mathbf{K}} &= \mathbf{R}^T \mathbf{K} \mathbf{R}, \\ \tilde{\mathbf{M}} &= \mathbf{R}^T \mathbf{M} \mathbf{R}, \end{aligned} \tag{4.5}$$

and the reduced system is finally solved. If there is damping, the reduced damping matrix is built using the same formula as for the mass and stiffness matrices above, with the same reduction matrix

R.

4.1 Guyan–Irons method

Static condensation is obtained by neglecting inertia in the condensed DOFs. The second equation of Equation 4.3 is written

$$\mathbf{K}_{CR}\mathbf{x}_R + \mathbf{K}_{CC}\mathbf{x}_C = \omega^2(\mathbf{M}_{CR}\mathbf{x}_R + \mathbf{M}_{CC}\mathbf{x}_C)\ . \tag{4.6}$$

Which, by neglecting the inertial terms, leads to

$$\mathbf{x}_C = -\mathbf{K}_{CC}^{-1}\mathbf{K}_{RC}\mathbf{x}_R\ . \tag{4.7}$$

The reduction matrix is therefore

$$\mathbf{R} = \begin{bmatrix} \mathbf{I} \\ -\mathbf{K}_{CC}^{-1}\mathbf{K}_{CR} \end{bmatrix}. \tag{4.8}$$

4.1.1 Results of the modal analysis

Table C shows the natural frequencies obtained when applying the Guyan-Irons reduction method by keeping only the translational DOFs associated to the three nodes in purple in Figure 32.

Table C: Natural frequencies of the first six modes computed on the full model and on the Guyan–Irons reduced model.

Model	**Mode 1**	**Mode 2**	**Mode 3**	**Mode 4**	**Mode 5**	**Mode 6**
Full model [Hz]	2.573	6.844	10.856	12.916	13.866	14.877
Guyan–Irons [Hz]	2.670	7.737	13.319	14.882	41.918	78.475
Relative error [%]	3.79	13.05	22.69	15.22	202.29	427.48

The natural frequencies of the reduced model are consistently higher than those of the full model. This is expected because Guyan–Irons static condensation removes inertia contributions associated with the eliminated DOFs, virtually reducing the mass of the system. Since the natural frequencies scale as

$$f \sim \sqrt{\frac{k}{m}}, \tag{4.9}$$

a reduction of mass leads to an overestimation of the frequencies, or equivalently adding effective stiffness to the system.

The effect becomes more pronounced for higher modes, as inertia forces scale like ω^2. As a consequence, modes associated to higher frequencies rely more on dynamic contributions neglected in the static condensation, leading to greater errors.

This can be further observed in the MAC (Modal Assurance Criterion) matrix – quantifying the correlation between 2 mode shapes – showed in Figure 33.

The first two modes show strong correlations, however there is a strong decrease from third mode and correlation becomes negligible from forth mode and above.
This confirms that the Guyan-Irons method can be reliable for low-frequency modes, but totally fails to reproduce higher-frequency modes, where inertia terms become more significant.

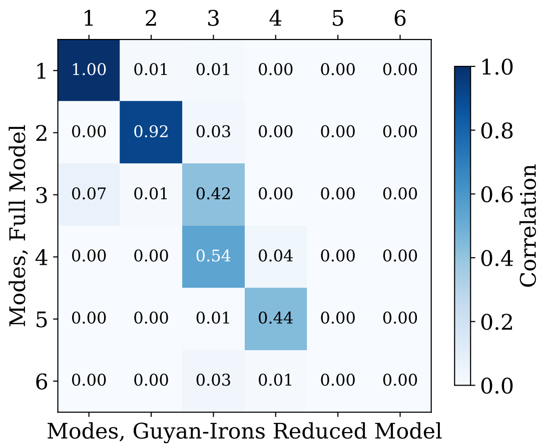


Figure 33: MAC matrix comparing the Guyan–Irons reduced modes with the full-model modes. Values close to 1 indicate strong correlation.

4.2 Craig-Bampton method

Craig–Bampton reduction retains only the *boundary* (B) DOFs and condenses all *internal* (I) DOFs. Their influence is captured through:

- Static modes : the static deformation of the internal DOFs induced by unit boundary displacements,
- the fixed–boundary eigenmodes of the internal subsystem.

The static influence of the internal DOFs is given by the second equation of Equation 4.3 in which the inertial terms are dropped (static analysis), leading to

$$\mathbf{x}_I^S = -\mathbf{K}_{II}^{-1}\mathbf{K}_{IB}\mathbf{x}_B. \tag{4.10}$$

And the eigenmodes are obtained from the second equation of Equation 4.3, where $\mathbf{x}_B = \mathbf{0}$:

$$\mathbf{K}_{II}\mathbf{x}_I = \omega^2\mathbf{M}_{II}\mathbf{x}_I. \tag{4.11}$$

From which the first n eigenmodes $\mathbf{x}_I^{(i)}$ are assembled into

$$\Phi_n = \begin{bmatrix}\mathbf{x}_I^{(1)} & \cdots & \mathbf{x}_I^{(n)}\end{bmatrix}. \tag{4.12}$$

The Craig–Bampton transformation is finally given by

$$\mathbf{x} = \mathbf{R}\begin{bmatrix}\mathbf{x}_B \\ \eta\end{bmatrix} = \begin{bmatrix}\mathbf{I} & \mathbf{0} \\ \underbrace{-\mathbf{K}_{II}^{-1}\mathbf{K}_{IB}}_{\text{static influence}} & \underbrace{\Phi_n}_{\text{dynamic influence}}\end{bmatrix}\begin{bmatrix}\mathbf{x}_B \\ \eta\end{bmatrix} \tag{4.13}$$

where η is a vector of generalized coordinates representing the influence of the first n modes of the internal structure.
Compared with Guyan–Irons, this method includes the dynamic contribution of the condensed DOFs and allows tuning the accuracy through the number of retained internal modes.

4.2.1 Results of the modal analysis

As the accuracy of the reduction can be tuned using the number of internal modes retained, a convergence study was performed to determine the number of internal modes required to obtain a maximum relative frequency error of 1% on the first six modes. The results in Figure 34 show that convergence is rapid: five internal modes yield errors of the order of 1%, and seven modes are required for the maximum relative error on the first six natural frequencies to be below 1%.

The reduced-system frequencies are slightly higher than those of the full model (Table D), which is expected because the inertia contributions of high-frequency internal modes are not retained. The MAC matrix in Figure 35 indicates an almost perfect correlation between the Craig–Bampton and full-model mode shapes; the correlation decreases only slighly for the highest-frequency modes considered.

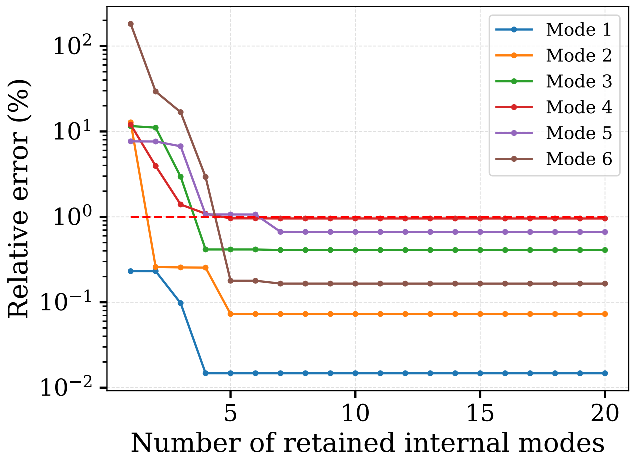


Figure 34: Relative error of the Craig-Bampton method on the six first natural frequencies as a function of the number of retained internal modes. 7 modes are required for the maximum relative error to be below 1 %.

Table D: Natural frequencies of the first six modes computed on the full model and on the Craig–Bampton reduced model with 7 internal modes.

Model	**Mode 1**	**Mode 2**	**Mode 3**	**Mode 4**	**Mode 5**	**Mode 6**
Full model [Hz]	2.5730	6.8440	10.8558	12.9163	13.8666	14.8772
Craig–Bampton [Hz]	2.5734	6.8490	10.9002	13.1000	13.9592	14.9018
Relative error [%]	0.015	0.073	0.409	0.957	0.668	0.165

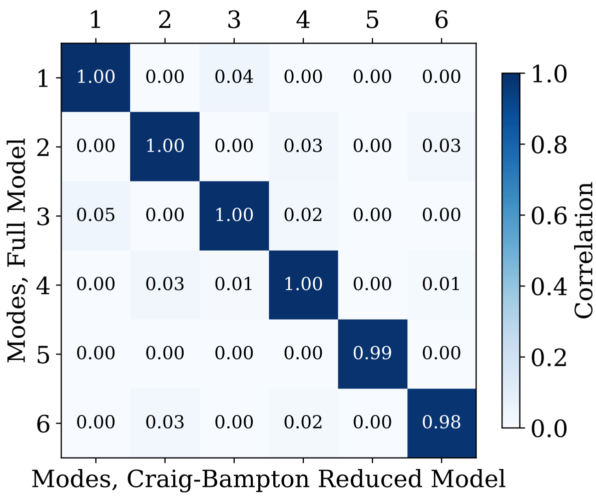


Figure 35: MAC Matrix of the Craig-Bampton modes correlation with the full model modes, using $n = 7$ internal modes.

4.2.2 Comparison with Guyan-Irons method

Table E: CPU time required to solve the eigenvalue problem, excluding the reduction-matrix computation. Values are mean $\pm$ standard deviation over 10 runs.

	Full model	Guyan–Irons	Craig–Bampton
Average CPU time [s]	1.07 ± 0.04	$(7 \pm 0.1) \times 10^{-4}$	$(1.2 \pm 0.2) \times 10^{-3}$

The comparison between the two reduction methods is based on both accuracy and computational cost[3]. A summary of the reduced-model errors is given in Figure 36, and the eigenvalue-solution CPU times for the full and reduced models are given in Table E.

Craig–Bampton consistently achieves errors one to two orders of magnitude lower than Guyan–Irons. This is expected: Craig–Bampton retains a set of internal dynamic modes, whereas Guyan–Irons is based purely on static modes and neglects dynamic effects of the condensed DOFs.

[3] All CPU times were measured on the same machine under identical conditions to eliminate setup dependence. Reported values are representative averages over multiple executions, as some variance remains.

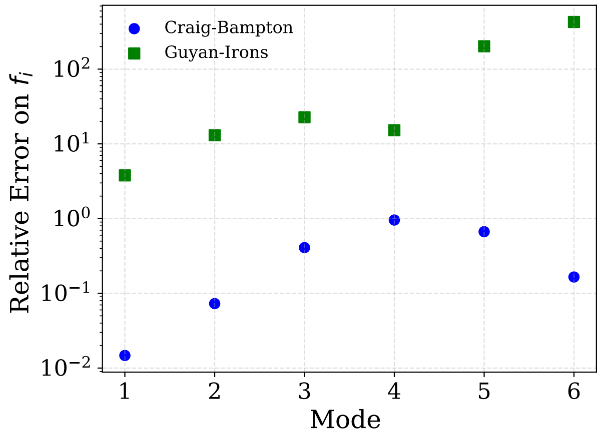


Figure 36: Relative errors on natural frequencies associated to the six first mode shapes of both reduction methods

The reduced eigenvalue systems are approximately three orders of magnitude cheaper to solve than the full-model system. The Craig–Bampton reduced system requires about 50% CPU time than Guyan–Irons due to the larger reduced basis, showing that accuracy comes with a cost. However this cost remains negligible compared to the full-model computation.

These timings exclude the computation of the reduction matrices – ~ 1 second for both methods– way larger than the cost of solving the reduced systems themselves. This cost is also greater for the Craig-Bampton method as it requires to solve the eigenvalues system for the internal subsystem – ~ 0.1 second from our measurements[4].

Overall, if excluding the computational time required to perform the reduction[5], the Craig-Bampton method clearly surpasses the Guyan-Irons method, with substantially smaller errors, with only a minor additional computational cost.
The reduced systems are both solved much faster than the full system.

4.3 Time Integration

To compare the reduction models on a transient time response, the case studied in section 3 is reused : a Y-axis parallel force of $7000\ N$ applied for $0.01\ s$ on the brown node in Figure 11. Time integrations are performed with parameters $h = 10^{-4}\ s$, $\gamma = 1/2$ and $\beta = 1/4$ so that the integration scheme is unconditionally stable and there is no high-frequency damping, and the periodicity error due to the integration scheme can be considered negligible compared to the errors of the reduction methods.

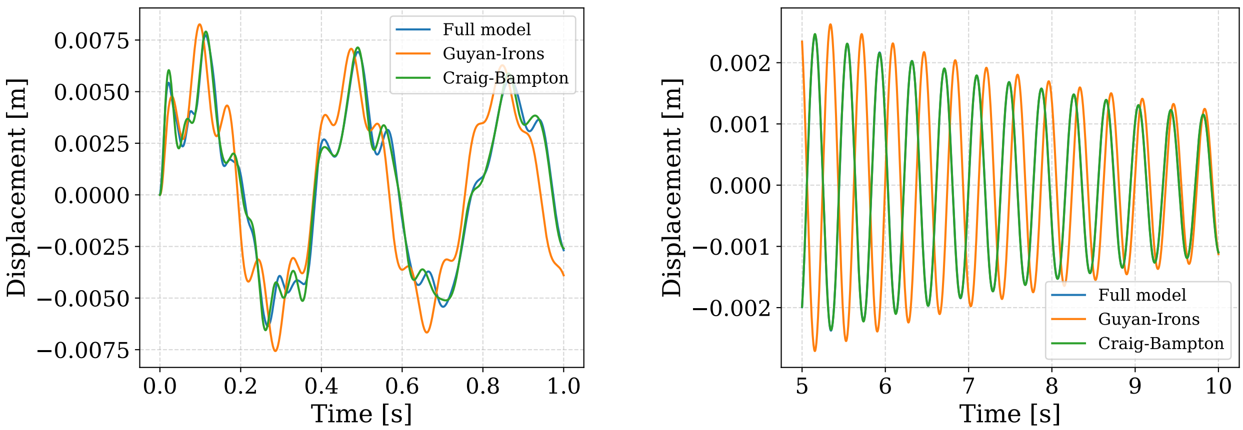
[4]But in theory should be similar to the CPU time required for solving the full system as the size of the full system and of the internal subsystem are approximately equal.

[5]If they are taken into account, the computational costs associated to the reduction processes are similar to the cost of solving the full system

4.3.1 Early Transient (0-1s)

In this initial phase, the response contains the full modal superposition, including high-frequency modes that decay rapidly due to damping. This interval is therefore useful to assess how accurately each reduced model reproduces the high-frequency dynamic content.

Figure 37a compares the full model with the two reduced models during this phase.



(a) Early Transient response

(b) Late Transient response

Figure 37: Comparison of the transient response at the excitation node for the full and reduced models. Newmark parameters : $\gamma = 0.5$, $\beta = 0.25$; time step $h = 10^{-4}$ s.

Table F: Relative error on the maximal magnitude of the time response for the reduced models.

Model reduction method	Guyan–Irons	Craig–Bampton
Peak amplitude error [%]	6.83	2.28

Both reduced models exhibit errors, but their nature differs. The Guyan–Irons model shows large errors in amplitude, frequency content, and overall waveform, reflecting the fact that the static condensation cannot represent inertia-dominated high-frequency modes. In contrast, the Craig–Bampton model preserves the general shape and frequency content of the full-model response; only a small amplitude error is observed.
This confirms that the Guyan–Irons reduction is unable to represent the high-frequency contributions accurately, whereas the Craig–Bampton model retains a coherent early-transient behavior.

For a quantitative comparison, the peak displacement provides a meaningful measure because it relates directly to potential structural damage. As reported in Table F, both models overestimate the maximum displacement. The Craig–Bampton model remains closer to the full-model result, but its error is still not negligible.

4.3.2 Late Transient (5-10s)

In this phase, all modes except the first one have been attenuated by structural damping. This allows to isolate and evaluate how accurately each reduced model represents the fundamental mode. Figure 37b compares the full model with the reduced models in reproducing the decay of this first mode. The Craig–Bampton method reproduces the full-model response with essentially no visible error. In contrast, the Guyan–Irons model exhibits an amplified displacement and a higher apparent natural frequency, consistent with the overestimation previously observed in its computed eigenvalues. This confirms again that static condensation provides a lower-quality approximation than Craig–Bampton, even for low-frequency modes where it performs best.

4.3.3 Frequency domain

To confirm the observations made above, the discrete Fourier Transform is applied to the time responses over the full interval up to $T = 10$ s. The result is shown in Figure 38. The overall frequency content of the Craig–Bampton model is nearly exact: the first natural frequency is captured accurately, and deviations appear only above approximately 12 Hz, where the amplitudes begin to slight visible errors while remaining close to the full-model spectrum.
In contrast, the Guyan–Irons model provides a reasonable representation of the first natural frequency—although its peak is overamplified and shifted to a slightly higher frequency—but the remainder of the spectrum is poorly reproduced. Higher natural frequencies are significantly shifted and their amplitudes differ significantly from the full-model values.

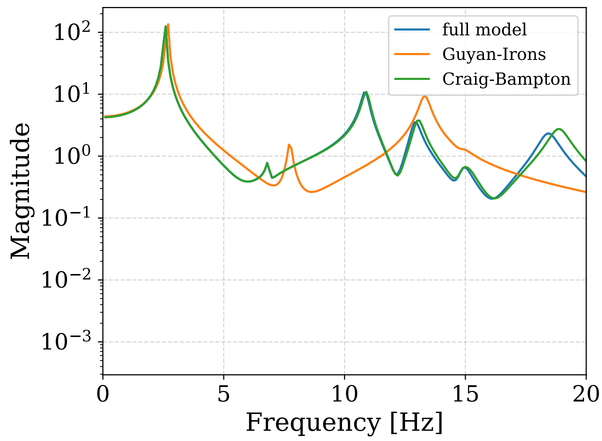


Figure 38: Comparison of the Discrete Fourier Transforms of the time response for the full and reduced models. Newmark parameters : $\gamma = 0.5$, $\beta = 0.25$; time step $h = 10^{-4}$ s and time period $t = [0; 10]$ s

4.3.4 Computational cost and model size

This section examines the computational benefit of performing model reduction.

Table G reports the CPU time required for 50 independent time-integration runs over $T = 1$ s with a time step $h = 10^{-4}$ s, together with the sizes of the three systems.

Table G: CPU time required to compute the time integration over $T = 1\,\mathrm{s}$ with time step $h = 10^{-4}\,\mathrm{s}$ and the system sizes. CPU times of the reduction process are excluded. Values are mean ± standard deviation over 50 runs.

	Full model	Guyan–Irons	Craig–Bampton
CPU time [s]	12.4 ± 0.6	0.13 ± 0.01	0.14 ± 0.01
System size [-]	1236	9	16

The model sizes are set by the desired level of accuracy: the full model and the Guyan–Irons model can only increase accuracy by adding physical DOFs, whereas the Craig–Bampton method can refine the approximation by retaining more internal modes for a fixed number of boundary DOFs. The full model requires however has a much larger size, here by roughly a factor 100, hence why a reduction is performed.

This difference in size translates directly into increased computational cost too, since both time integration and eigenvalue computations essentially consist in solving algebraic systems.

The observed measurements show an approximately linear increase of the CPU time with an increase of the system size n, not a theoretical $O(n^2)$ or $O(n^3)$ behavior –depending on the method used to solve the system. Despite this quasi-linear behavior, the difference of computational cost in between reduced models and full model solving is significant, showing the benefits of using the reduced models for practical applications.

When comparing the two reduced models, the additional cost associated with the Craig–Bampton system is marginal—-less than 10%—-and is negligible relative to the significant accuracy improvement that this method provides.

A AI charter

Generative Artificial Intelligence tools, such as OpenAI's chatGPT and Google Gemini, were used as

- a helping and debugging tool for the scripts for plots generation.
- a debugging and documentation tool for LaTeX report writing.

B Computation of δ_{rel}

$$\delta_{\text{rel}}^{N \to N+1} = \frac{|f_{N+1} - f_N|}{f_{N+1}} \times 100\% \tag{2.1}$$

where f_i and f_{i+1} are the computed values (e.g., natural frequencies) for two successive meshes.

C Elementary matrices expressions

Using the following partitioning,

$$\mathbf{d} = \{\mathbf{d}_1, \mathbf{d}_2\}^T, \quad \text{where } \mathbf{d}_i = \{u_{ix}, u_{iy}, u_{iz}, \theta_{ix}, \theta_{iy}, \theta_{iz}\}^T \tag{3.1}$$

Euler-Bernoulli beam theory gives the following elementary matrices, where $r = I/A$ the

$$\boldsymbol{K}_{el} = \begin{bmatrix}
\frac{EA}{l} & & & & & & & & & & & \\
0 & \frac{12EI_z}{l^3} & & & & & & & & & & \\
0 & 0 & \frac{12EI_y}{l^3} & & & & & & & & & \\
0 & 0 & 0 & \frac{GJ_x}{l} & & & & & & & & \\
0 & 0 & \frac{-6EI_y}{l^2} & 0 & \frac{4EI_y}{l} & & & & SYM & & & \\
0 & \frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{4EI_z}{l} & & & & & & \\
\frac{-EA}{l} & 0 & 0 & 0 & 0 & 0 & \frac{EA}{l} & & & & & \\
0 & \frac{-12EI_z}{l^3} & 0 & 0 & 0 & \frac{-6EI_z}{l^2} & 0 & \frac{12EI_z}{l^3} & & & & \\
0 & 0 & \frac{-12EI_y}{l^3} & 0 & \frac{6EI_y}{l^2} & 0 & 0 & 0 & \frac{12EI_y}{l^3} & & & \\
0 & 0 & 0 & \frac{-GJ_x}{l} & 0 & 0 & 0 & 0 & 0 & \frac{GJ_x}{l} & & \\
0 & 0 & \frac{-6EI_y}{l^2} & 0 & \frac{2EI_y}{l} & 0 & 0 & 0 & \frac{6EI_y}{l^2} & 0 & \frac{4EI_y}{l} & \\
0 & \frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{2EI_z}{l} & 0 & \frac{-6EI_z}{l^2} & 0 & 0 & 0 & \frac{4EI_z}{l}
\end{bmatrix}$$

Figure 39: Euler-Bernoulli beam element stiffness matrix

Timoshenko Beam element Using the same partitioning as above The stiffness matrix of the Timoshenko beam element is written as:

$$\mathbf{K}_T = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix} \tag{3.2}$$

Introducting shear-dependent terms Φ_y and Φ_z, and β_y and β_z:

$$
\boldsymbol{M}_{el} = \rho A l \begin{bmatrix}
\frac{1}{3} & & & & & & & & & & & \\
0 & \frac{13}{35} & & & & & & & & & & \\
0 & 0 & \frac{13}{35} & & & & & & & & & \\
0 & 0 & 0 & \frac{r^2}{3} & & & & & & & & \\
0 & 0 & \frac{-11l}{210} & 0 & \frac{l^2}{105} & & & & SYM & & & \\
0 & \frac{11l}{210} & 0 & 0 & 0 & \frac{l^2}{105} & & & & & & \\
\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & & & & & \\
0 & \frac{9}{70} & 0 & 0 & 0 & \frac{13l}{420} & 0 & \frac{13}{35} & & & & \\
0 & 0 & \frac{9}{70} & 0 & \frac{-13l}{420} & 0 & 0 & 0 & \frac{13}{35} & & & \\
0 & 0 & 0 & \frac{r^2}{6} & 0 & 0 & 0 & 0 & 0 & \frac{r^2}{3} & & \\
0 & 0 & \frac{13l}{420} & 0 & \frac{-l^2}{140} & 0 & 0 & 0 & \frac{11l}{210} & 0 & \frac{l^2}{105} & \\
0 & \frac{-13l}{420} & 0 & 0 & 0 & \frac{-l^2}{140} & 0 & \frac{-11l}{210} & 0 & 0 & 0 & \frac{l^2}{105}
\end{bmatrix}
$$

Figure 40: Euler-Bernoulli beam element mass matrix

$$
\beta_y = 1 + \Phi_y, \quad \beta_z = 1 + \Phi_z,
$$
$$
\Phi_y = \frac{12EI_y}{L^2AG}, \quad \Phi_z = \frac{12EI_z}{L^2AG},
$$

The four sub-matrices write as

$$
\mathbf{K}_{11} = \begin{pmatrix}
\frac{EA}{L} & & & & & \\
0 & \frac{12EI_z}{L^3\beta_z} & & SYM & & \\
0 & 0 & \frac{12EI_y}{L^3\beta_y} & & & \\
0 & 0 & 0 & \frac{GJ}{L} & & \\
0 & 0 & -\frac{6EI_y}{L^2\beta_y} & 0 & \frac{(4+\Phi_y)EI_y}{L\beta_y} & \\
0 & \frac{6EI_z}{L^2\beta_z} & 0 & 0 & 0 & \frac{(4+\Phi_z)EI_z}{L\beta_z}
\end{pmatrix} \tag{3.3}
$$

$$
\mathbf{K}_{12} = \begin{pmatrix}
-\frac{EA}{L} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{12EI_z}{L^3\beta_z} & 0 & 0 & 0 & \frac{6EI_z}{L^2\beta_z} \\
0 & 0 & -\frac{12EI_y}{L^3\beta_y} & 0 & -\frac{6EI_y}{L^2\beta_y} & 0 \\
0 & 0 & 0 & -\frac{GJ}{L} & 0 & 0 \\
0 & 0 & \frac{6EI_y}{L^2\beta_y} & 0 & \frac{(2-\Phi_y)EI_y}{L\beta_y} & 0 \\
0 & -\frac{6EI_z}{L^2\beta_z} & 0 & 0 & 0 & \frac{(2-\Phi_z)EI_z}{L\beta_z}
\end{pmatrix} \tag{3.4}
$$

$$
\mathbf{K}_{21} = \mathbf{K}_{12}^T \tag{3.5}
$$

$$
\mathbf{K}_{22} = \begin{pmatrix}
\frac{EA}{L} & & & & & \\
0 & \frac{12EI_z}{L^3\beta_z} & & SYM & & \\
0 & 0 & \frac{12EI_y}{L^3\beta_y} & & & \\
0 & 0 & 0 & \frac{GJ}{L} & & \\
0 & 0 & \frac{6EI_y}{L^2\beta_y} & 0 & \frac{(4+\Phi_y)EI_y}{L\beta_y} & \\
0 & -\frac{6EI_z}{L^2\beta_z} & 0 & 0 & 0 & \frac{(4+\Phi_z)EI_z}{L\beta_z}
\end{pmatrix} \tag{3.6}
$$

Note that when Φ_y and Φ_z are 0, the stiffness matrix is the same as the one of Euler-Bernoulli theory

Using the same partitioning, the elementaty mass matrix for Timoshenko beam elements writes as

$$\mathbf{M}_T = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} \tag{3.7}$$

where $\mathbf{M}_{22} = \mathbf{M}_{11}$ and $\mathbf{M}_{21} = \mathbf{M}_{12}^T$.

$$\mathbf{M}_{11} = \frac{\rho A L}{420} \begin{pmatrix} 140 & & & & & \\ 0 & 156 & & & SYM & \\ 0 & 0 & 156 & & & \\ 0 & 0 & 0 & 140\frac{I_x}{A} & & \\ 0 & 0 & -22L & 0 & \left(4L^2 + 140\frac{I_y}{A}\right) & \\ 0 & 22L & 0 & 0 & 0 & \left(4L^2 + 140\frac{I_z}{A}\right) \end{pmatrix} \tag{3.8}$$

$$\mathbf{M}_{12} = \frac{\rho A L}{420} \begin{pmatrix} 70 & & & & & \\ 0 & 54 & & & SYM & \\ 0 & 0 & 54 & & & \\ 0 & 0 & 0 & 70\frac{I_x}{A} & & \\ 0 & 0 & 13L & 0 & \left(-3L^2 - 70\frac{I_y}{A}\right) & \\ 0 & -13L & 0 & 0 & 0 & \left(-3L^2 - 70\frac{I_z}{A}\right) \end{pmatrix} \tag{3.9}$$